

Beyond Publication Bias: Characterizing and Understanding Missing Results in Economics

Fernando Hoces de la Guardia^{*†} Edward Miguel[‡]
Gufran Pathan[†] Akash Shaji[†] Viviane Helena Silva da Rocha[§]
Erik Ø. Sørensen[§] Bertil Tungodden[§]

October 1, 2026
Job Market Paper: Latest Version

Abstract

What scientists choose not to publish can be as important as what they do publish. This study introduces a novel measure of the reporting of research results in economics, and investigates the causes behind under-reporting. Across 5,048 pre-registered hypotheses from 317 studies on the American Economic Association RCT Registry that were encoded, we first show that only 42% of hypotheses, on average across studies, have publicly available results 8 to 9 years after pre-registration. Around 64.5% of these publicly available findings are null results (i.e., not statistically significant), contrary to widespread concerns about the non-reporting of null results. An RCT was carried out to test three plausible barriers to full reporting, namely, a lack of *awareness*, *engagement*, or *resources*. An intervention in which our team engaged directly with the authors' research and their under-reported findings increased the share of hypotheses with publicly available results by 6.2 percentage points (11% of the missing hypotheses), though most of this increase reflected results that already existed but were buried or previously observed; genuinely new estimates account for only 2.5 percentage points. Authors explained a further 18% of missing hypotheses as cases where the planned research never occurred; these explanations document why a result is absent rather than providing it, and although they do little for publication bias, they help limit research waste by creating a record of which planned studies were never carried out. Neither a light-touch "awareness" intervention nor a more expensive research-assistance intervention added to reporting, indicating that recovery required hands-on engagement. Together, the findings indicate that the majority of pre-registered hypotheses in economics remain unreported, with 42% still unaccounted for even after our most intensive arm; that under-reporting is more complex than a simple reluctance to report nulls; and that recovering missing results is difficult and labor-intensive rather than low-cost. We conclude that there is still need for stronger incentives and infrastructure to ensure that pre-registered scientific plans translate into publicly accessible evidence. Toward that end, this study provides prototypes of how standardized results reporting could be incorporated into economics.

^{*}Corresponding author: fhoces@gmail.com. We thank Katie Baker, Wilson King, Alix Schoback, Sarah Stillman, Michael Walker, and Johanna Weech for providing valuable feedback and assistance on this project, and to seminar audiences for valuable feedback. The study is registered on the AEA Registry, AEARCTR-0013755, <https://doi.org/10.1257/rct.13755-2.0>.

[†]University of California, Berkeley

[‡]University of California, Berkeley and NBER

[§]NHH Norwegian School of Economics

1 Introduction

At every stage of the research process, from data collection to publication, results that challenge prevailing theories or that simply fail to establish a robust relationship may be suppressed by disappointed researchers, skeptical referees, or editors worried about page counts. Such selective dissemination of findings can distort the body of research evidence and lead to a biased scientific consensus, a phenomenon Rosenthal (1979) termed the “file drawer problem.” The published record will tend to over-represent statistically significant results, and estimates of important effects will generally be biased. The failure to report to the community that certain hypotheses are difficult to test may also lead to duplication of effort and the waste of scarce research resources.

While the *existence* of the file drawer and its associated problems seems intuitive and difficult to reject, until recently the extent of under-reporting of research results has proven challenging to measure. A leading obstacle is that the set of hypotheses to be tested is typically unobservable, and possibly not even well defined *ex ante*. However, the recent emergence of study registries across multiple disciplines, including in economics, offers a new opportunity for studying the problem directly. By requiring researchers to specify hypotheses and analytical strategies before observing data, study registries make it possible to compare the intended versus realized reporting of results.

Building on this insight, this study leverages the American Economic Association (AEA) Randomized Controlled Trial (RCT) Registry to address two primary questions. First, how large is the file drawer problem among RCTs in economics? Second, what factors influence researchers’ decisions about which results to make publicly available? To address the first question, we developed a new method to standardize the encoding of hypotheses listed on the AEA Registry, and systematically locate and archive their corresponding results in the publicly available literature. To address the second, we carried out an RCT to causally identify the constraints that may prevent researchers from reporting results in a complete and transparent manner.

Study registries have their background in medicine as far back as the US National Cancer Act of 1971, which allowed for the establishment of an “...international cancer research data bank to collect, catalog, store, and disseminate insofar as feasible the results of cancer research...” (Congress, 1971, §407). While this did not start as a modern prospective registry — in which design, outcomes, and hypotheses were registered prior to data collection — it developed into the database used by Simes (1986) to show that, in reviewing a particular treatment of ovarian cancer, a literature search of all published trials provided quite different results from reviews of registered trials only, trials that were arguably less subject to the file drawer problem.¹ Prospective registration of studies (known as “pre-registration”) was embraced by the economics profession with the establishment of the AEA RCT Registry in 2013 and registration was made mandatory for publication in AEA journals in 2015. By 2015, about one thousand registered trials

¹Simes’ arguments in favor of a trials registry are similar to those that led to the establishment of large general registries in medicine such as `ClinicalTrials.gov`, which went online in 2000, and was followed by the 2004 statement from the International Committee of Medical Journal Editors that “[t]he ICMJE member journals will require, as a condition of consideration for publication, registration in a public trials registry” (De Angelis et al., 2004, p. 1250). See also Thibault and TARG Meta-Research Group (2023) for reviews of the completeness of reporting in medicine.

had started, and at the time of writing the number is over 11,000. For each registration, research teams fill out a form outlining the study’s experimental design, the main and secondary outcomes, population studied, sampling strategy, randomization mechanism, and sample size. About a third have also submitted a pre-analysis plan (PAP) in which they lay out their approach in more detail. The core features of each registration are publicly available and time-stamped (AEA RCT Registry, 2025).

In this study, a schema was developed for encoding hypotheses directly from AEA RCT study registrations to establish the extent of the file drawer problem. In addition to collecting basic study and investigator information (largely compatible with DDI Alliance (2021)), the schema incorporates the *outcomes*, *interventions*, and *arms* structure proposed by Cavanagh et al. (2023), and introduces new fields to encode hypotheses in greater detail. The schema was applied to pre-registrations in the AEA Registry between 2015 and 2017. Although some elements can be directly extracted from the registry database, substantial manual effort was often required to read and interpret the registrations; in several cases, it was also necessary to contact study investigators to obtain PAPs that are only available upon request. After encoding all main hypotheses for each registration, the research team conducted an extensive search for all publicly available study outputs, including published articles or book chapters, working papers, reports, or presentations. These outputs were reviewed in order to identify results corresponding to the encoded hypotheses. For each matched hypothesis, we recorded estimated effect sizes, standard errors, and *p*-values, as well as metadata (such as the location of the result within the materials), whether the reported results correspond exactly to the registered hypotheses, or if there were some modifications (i.e. slight or substantial variations in definitions of population, outcomes, arms or in the econometric model), whether results were reported in the main text versus the appendix, and a subjective assessment of how difficult it was to locate the corresponding result.

After encoding the $N=317$ pre-registered studies, individual studies were then randomly assigned to one of four study arms. In the **Control** arm, registrations were encoded but study authors were not contacted. In the other three arms, authors were contacted by email with different interventions designed to promote more complete reporting of the pre-registered hypotheses, including via the uploading of any previously unreported results to the AEA Registry. Most recipients of the email did engage with it: about 89.1% clicked on a link in the email and about 65.4% responded to our team. A standardized form, the *Results Report*, was also developed to facilitate the reporting of results from pre-registered hypotheses.

In the **Awareness** arm, study authors were emailed a message emphasizing the value to the scientific community of full reporting of results, along with an attached blank Results Report. In the **Engagement** arm, authors received the same email message, but in addition their respective Results Reports were pre-populated with the information obtained during encoding, namely, the results that we were able to locate as well as a listing of those hypotheses for which no results had been located. In the **Resources** arm, authors received the previous interventions and additionally an offer of 30 hours of research assistance (or an equivalent USD 1,500 in a small research grant) to support the completion of their Results Report. Six months after contacting the authors, the research team conducted a follow-up search for any new publicly available study outputs (i.e., papers or reports) and encoded them as post-intervention outcomes.

In parallel with the encoding of the studies, a survey of forecasts among experts was conducted using the Social Science Prediction Platform (SSPP). The survey elicited expert beliefs about the degree of publication bias and selective reporting in economics in a way closely tied to this study’s encoding of pre-registered hypotheses and results.

The first set of main findings in this study relate to the extent of publication bias among pre-registered economics studies. We show that no research outputs – i.e., no published articles, working papers, or chapters/reports – could be located for 19.2% of the studies pre-registered on the AEA Registry. It is worth noting that this is similar to a result in Franco et al. (2014), who examined papers resulting from a subset of National Science Foundation (NSF) grant applications. However, that study and other earlier work largely stop at the study level and do not carry out a hypothesis-level analysis as we are able to do in this study, and which we view as a major strength.² This approach allows us to expand our understanding of selective reporting. Although we found that around 80.8% of pre-registered studies produce some research output, results are publicly available for only 42% of hypotheses. We view generating this figure as one of the main contributions of the current study. In other words, among the 80.8% of studies with some research output, results are reported for only roughly half of the pre-specified hypotheses. The distribution of results reporting is also noteworthy: we were able to locate all pre-specified hypotheses for roughly 20.2% of the pre-registrations while 48.3% of all studies report only a subset of pre-specified hypotheses. While the data indicate that there is substantial under-reporting of research findings in economics, note that the 42% rate of pre-specified results being reported is considerably higher than the level predicted in the expert forecasts.

The RCT yields a second set of main results.³ The data indicate that the Engagement and Resources intervention arms both led to significantly higher reporting of pre-specified results than in the Control group, by 6.9 percentage points and 5.5 percentage points, respectively (although the difference between them is not statistically significant), whereas the Awareness arm alone did not increase reporting. Recall that 42% of pre-specified hypotheses were publicly available when we searched for them; thus the roughly 6.2 percentage point average increase in reporting across the Engagement and Resources arms means that results became available for roughly 11% of those not reporting (i.e., on a base of 58% of missing results). Most of this increase, however, reflected results that already existed but were hard to find or had already been observed; only 2.5 percentage points (about 4% of the file drawer) were genuinely new estimates absent anywhere at baseline. The great majority of the file drawer thus remains untouched.

One obvious explanation for the large share of “missing” unreported findings is the possibility that certain research was never carried out, and the data we assemble – based on email exchanges with authors in the three intervention arms – also generates insights into the extent of this issue. The data show that either interventions did not occur or data

²van den Akker et al. (2023) studied a sample of psychology studies and found that roughly half contained omitted hypotheses. Their sample differs from ours along several dimensions: they focus on published and prize-winning papers from a “Preregistration Challenge”, and most were laboratory or survey experiments in psychology. Ofosu and Posner (2023) examined PAPs on the AEA Registry and found that only 55% were associated with a published paper, lower than the 80.8% we document. This lower figure may reflect that their observation window was far shorter – at 2–6 years after registration (at which time some studies may still have been ongoing) – than our 8–9 year follow-up period.

³The RCT was pre-registered on the AEA RCT Registry, at AEARCTR-0013755.

collection was not carried out for 10% of pre-registered hypotheses. Thus in those cases where the pre-specified research could have been conducted, the actual rate of reported hypotheses is moderately higher, at approximately 47%, or roughly half. There is little prior research on interventions aimed at increasing reporting of pre-specified results, with Cavanagh et al. (2024) a notable exception.⁴

A second set of issues that we pre-registered relates to the proportion of null, or not statistically significant, results reported. Both the original argument by Rosenthal (1979) and the empirical findings of Franco et al. (2014) suggest that unpublished studies are disproportionately more likely to contain null results. Professional researchers also believe there is a strong “null result penalty” in the publication process (Chopra et al., 2024). Yet we find that the share of reported pre-specified tests that are null is remarkably high, at 64.5% of reported results (at the conventional 95% confidence threshold). This is far higher than the share predicted in the SSPP expert forecasts we collected for this study (which was only 37%, roughly half of what we find). And these null results are not evenly spread out across studies: for about 30% of the pre-registered studies, over 90% of the publicly reported results are non-significant.

The additional reporting of results induced by the RCT arms (the Engagement and Resources arms) also allows us to test whether these “new” findings differ from those originally reported, but we find that the share of null results is – perhaps surprisingly – the same in both groups of findings. In other words, the marginal estimates that are uncovered through the RCT interventions are no more likely to be null results. This pattern argues against a simple view that economics researchers are systematically suppressing null findings from their research outputs, at least among those studies pre-registered on the AEA Registry that responded to the interventions. It remains an open question in our view (and one that is difficult to rigorously test) whether the existence of the AEA Registry itself is contributing to improved reporting of null findings, although it is certainly consistent with these patterns.

The findings of this study contribute to at least two literatures in economics and the social sciences. The first is the body of work documenting and analyzing publication bias in economics research. Andrews and Kasy (2019) model the likelihood of a result being published as a decreasing function of its p -value. Others have emphasized that even the p -value of a result itself might be subject to manipulation if researchers have sufficient flexibility (Simmons et al., 2011). Brodeur et al. (2016, 2020) show that there is bunching in the distribution of test statistics around conventional significance cutoffs that is arguably evidence of p -value manipulation. In a related study, Brodeur et al. (2024) examine the effect of pre-registration on such bunching of test statistics and show that pre-registration does not materially affect bunching unless there is also a PAP available, as its existence presumably provides more restrictions on the analysis.

A second literature to which this paper aims to contribute relates to the process of evidence aggregation in economics and other social sciences. This work includes the development of evidence gap maps that visualize the availability of studies across combinations of outcomes and interventions (Snijlsteit et al., 2016). Other efforts involve

⁴They conducted an RCT testing different nudge-style emails to authors with AEA RCT registrations, encouraging them to fill in post-trial information. They found only a 1 percentage point increase in reporting, with no meaningful differences across message types. Note that the interventions carried out in the current study are far more involved.

related data collection and standardization initiatives. For example, Vivalt (2020) uses crowd-sourcing to encode estimates for nearly 200 development studies; Hendren and Sprung-Keyser (2020) undertook a large-scale effort for 133 US policies, developing a unified framework for standardizing costs and benefits; and the World Bank leads an initiative to standardize results from all international development impact evaluations (IDEAL Consortium, 2025). The standardization approach and the *Results Reports* introduced in the present study directly advance this evidence-aggregation agenda by providing a streamlined hypothesis-level tool for collecting, structuring and reporting key pre-specified findings.

The rest of the paper is structured as follows. Section 2 outlines the experimental design and sample in more detail, and gives a detailed account of the encoding procedures. Section 3 defines the outcomes, the empirical strategy, and main hypotheses. Section 4 presents the results, and Section 5 provides a discussion and concluding remarks.

2 Data Collection and Experimental Design

This section discusses main elements of the study in turn, namely, the data encoding and descriptive analysis (Section 2.1), and the RCT design and analysis (Section 2.2). It then discusses the collection of expert forecasts (Section 2.3).

2.1 Data Collection and Encoding Process

A sample of pre-registered RCTs was created using the AEA Registry. The sampling frame consisted of all registrations between January 2015 and May 2017 ($n = 913$). In this period, RCT study pre-registration became increasingly common in economics, and focusing on this period also allows for a follow-up time frame of 8–9 years to the time of writing to track whether study results had become public. After applying pre-specified study inclusion and exclusion rules, the analysis sample consists of 317 pre-registrations (Figure 1). These rules include retaining only the earliest AEA registration for each principal investigator (–152), removing trials that were not genuinely pre-registered, i.e. registered only after the intervention or data collection had already been completed (–350), or that were insufficiently informative (–52), and excluding studies authored by this project’s own research team (–7); a further 35 eligible control studies were randomly set aside for possible later encoding.

A standardized procedure was then developed to encode both pre-registrations and study outputs, drawing on emerging taxonomies for RCT metadata in the social sciences (Cavanagh et al., 2023). This process involved two stages. First, each pre-registration and any associated PAPs were translated into a structured dataset of hypotheses (which we call G_0 , where the subscript denotes that this is the registration – no results). Second, all public outputs later linked to each registration, including published journal articles, working papers, and policy reports, were systematically searched, and the available results for each of the pre-registered hypotheses were recorded in a second structured dataset (which we call G_1 , denoting the follow-up results). Information from the registrations in G_0 and the subsequent research results in G_1 were encoded from July 2023

to August 2025, with follow-up encoding of newly available results extending through February 2026, ensuring that each study had at least 8 years for results to emerge.

[FIGURE 1 HERE]

The encoding of each registration (G_0) began by identifying the population(s) of interest in the study, which entailed identifying a country, a sample size (when possible), the unit of randomization, and a short description of the population of interest (e.g. “small business owners in city X”).⁵ The second element recorded were the main outcomes. There was considerable variation in the degree of detail provided by the authors for this field, including for how clearly the distinction was made between primary versus secondary outcomes.⁶ The most common sources of ambiguity were vaguely defined outcomes (e.g. ‘health’, ‘income’, ‘education’, ‘spending’, ‘trust’ rather than specific measures based on clearly noted questions or sources), under-specification on how any indices that combine multiple measures were to be constructed (i.e., simple averages, weighted averages, or something else), and discussion of whether specific measures or indices (or both) were considered primary outcomes.

Third, study “interventions” were encoded, and there was again considerable ambiguity in how these were described. The study “arms” were then specified based on interventions, where multiple interventions were sometimes combined into a single arm; for instance, there may have been two different microfinance interventions but the main test might have pooled both of these into a single arm and compared it to the control group. At this point, the potential for ambiguity and complexity increases substantially, as with more than two study interventions (counting control as one intervention) – which occurs in 65.9% of the studies in the sample – the number of possible combinations in assigning interventions to arms grows rapidly. Unfortunately, most registrations fail to clearly and explicitly define their arms. In this case, the encoders were instructed to define the arms with the simplest and most intuitive design possible or with a design that most closely addresses the research questions stated in the registration.

With all these elements in place, encoders proceeded to record each pre-registered hypothesis in a standardized fashion by recording the following components: the arms to contrast, the statistical tests to carry out, an alternative hypothesis, and the statistical estimation method (including standard error estimation). Finally, each hypothesis was classified into one of four categories, which were common examples of contrasts found in these pre-registrations: (1) a simple difference between conditional expectations, (2) a double difference between conditional expectations, (3) a joint hypothesis test (for instance, across multiple study arms), and (4) a final category that includes any other type of hypothesis that did not fit neatly into any of the previous categories, for instance, certain types of structural econometric estimation, although in practice this was relatively rare (as discussed below). Whenever judgment calls were required during encoding any

⁵This is already defined in a field in the AEA Registry, and due to its simplicity, it should already be more standardized and thus easier to encode. However, we found that data on the study country was not always recorded in a uniform fashion, requiring encoders to search for it in other fields or documents. This highlights how encoding even relatively simple fields requires some interpretation.

⁶Other authors have made similar observations, including Abrams et al. (2021); Ofosu and Posner (2023) and Leight et al. (2025).

of these elements, a record was created with the rationale and precedent (if any) behind the judgment call.⁷

All available information from the fields in the AEA registry and any PAPs found on the registry or elsewhere online were utilized to encode the dataset of pre-registered hypotheses (G_0). This includes all main hypotheses and any primary heterogeneity analyses. Only causal hypotheses based on the experimental designs were included, i.e., any descriptive or non-randomization-related tests were not included. Heterogeneity tests are included only when the pre-registration provides clear motivation and specification of subgroups or dimensions of interest. Relatively vague or exploratory mentions of subgroup analysis, or cases lacking sufficient detail (e.g., no detailed subgroup definitions), are thus not included in G_0 .

In the main analysis sample of 317 pre-registrations, after this encoding process there are a total of 3,731 main hypotheses and 1,317 primary heterogeneity tests in G_0 , for a total of 5,048 hypotheses (again see Figure 1).

The second stage of data collection involved identifying and recording the publicly available results corresponding to each pre-registered hypothesis recorded during the first stage, in what we call the G_1 data. For each hypothesis recorded in G_0 , a systematic online search was conducted across potential sources of dissemination, including published journal articles, working papers, policy reports, authors' websites, and any other publicly accessible materials. This process enabled the construction of a hypothesis-level database linking each pre-registered hypothesis to any related public available results. When a result was found, it was matched to its corresponding pre-registered hypothesis and recorded in a standardized fashion.

The process captured information on the availability of the result, as well as whether there were discrepancies identified between the originally specified version of the hypothesis, the outcomes, and treatment arms, versus what was found (and if so, whether a justification was offered for the modification). Fields were also recorded for each hypothesis-level result, including the coefficient estimate, p -value, standard error, statistical significance, sample size, source location (i.e., the article table in which it was found), whether the result was reported only in an appendix, and the effort required to find and extract the information (which was subjectively classified as easy, medium, or hard). If no result corresponding to the pre-registered hypothesis could be located in any of the materials, the hypothesis was coded as '*missing*'. If only qualitative evidence of the result was available (e.g., whether it was null, mixed, or strong), the hypothesis was coded as '*partially found*'; this is sometimes the case in practice if authors note in the text that a certain result was not statistically significant but do not present the detailed estimate or standard error, and other similar cases.

Given the unstructured nature of the source material (i.e., mainly published articles and working papers), extensive quality control was implemented to ensure consistent encoding. In a pilot, two principal investigators jointly coded five studies with the research assistants who later carried out most of the encoding. This collaborative approach was chosen instead of independent double-coding due to the high cost of encoding: we

⁷Specifically, an XML schema was developed to encode G_0 in a consistent manner, extending the fields defined in DDI Alliance (2021) and Cavanagh et al. (2023). The schema is available at <https://github.com/ErikOSorensen/rgtools>.

estimate that it takes approximately 11 hours per study on average to carry out the encoding process described above⁸. There is also evidence from related projects of significant initial differences in encoding reliability that later required senior researcher intervention (van den Akker et al., 2023), something that we attempted to address up front with PI involvement in initial studies, in order to develop standards and precedents for later studies. The first 10 studies of the study sample were then double-encoded, and initial results revealed considerable discrepancies, often stemming from different starting points (e.g., encoders using different versions of a pre-registration). To resolve these and other ambiguous cases that arose, a system of tracking encoding issues via GitHub and a full review of the first 100 encodings by one of the PIs was established. All lessons learned and resolutions were documented to guide future encoding decisions by the study team, and also by any other scholars who wish to replicate this approach. Over time, intensive collaboration across team members combined with daily monitoring by PIs led to qualitatively and quantitatively similar encoding of studies across research assistants⁹.

Table 1 presents descriptive statistics on the pre-registrations encoded in G_0 , including the central tendency (the mean and median) and distribution (quantiles) across multiple characteristics. At the study level, 40% of pre-registrations have a PAP (found either on the AEA Registry or elsewhere), and 65% of study populations are in low- and middle-income countries (LMICs), which can be broadly thought of as the field of development economics. Only 12% of studies specify heterogeneity analyses in sufficient detail to be coded as primary, as discussed above.

[TABLE 1 HERE]

The median study contains 9 hypotheses, while the mean is nearly 16, indicating a highly skewed distribution with a relatively small number of studies pre-registering many hypotheses. Each study typically describes around 5 main outcomes, with an average of 4 interventions and 5 experimental arms (including any control arms); recall that interventions can be combined in various ways into treatment arms, and there may be multiple control groups. On average across studies, only 31% of hypotheses were coded as ‘*detailed*’, meaning that the registration provided sufficient information for encoding without additional judgment calls by the encoder. This share differs sharply by study type: among studies with a PAP, 65% of hypotheses are considered detailed, compared to only 9% among those without a PAP. This pattern underscores the potential importance of PAPs in constraining researcher discretion and clarifying pre-registered analytical choices. Limited evidence in the literature suggests PAPs may have an effect on reducing the bunching of p-values around traditional statistical thresholds¹⁰ (Brodeur et al., 2024).

⁸Additional details on the encoding process are presented in Section A of the Appendix.

⁹When the descriptive results presented here are separated by encoder, the two main encoders (that account for 90.2% of the encoded studies) show remarkable similarities across most observable characteristics. The third encoder presents more differences in the encodings, but this could be due to sampling variation. See Tables A2 and A3 in the Appendix

¹⁰Limited in the sense that it compares a sample of around 20 papers, includes all p-values (around 60 per paper) related to results, does not observe original hypotheses, and defines publication bias in a narrow way (bunching around statistical threshold) on a sample (RCTs in Economics) for which previous research had shown small or no bunching (Brodeur et al., 2020)

Finally, 89.1% of hypotheses can be expressed as simple differences (e.g., between one treatment arm vs. control, or between two treatment arms) and 4.5% as a double difference, while only 6.4% fall into more complex categories. Thus most pre-registered studies on the AEA Registry during the 2015 to 2017 period featured research designs that can be expressed as direct experimental comparisons rather than, for instance, structural econometric estimates.

These design dimensions also help explain why the number of pre-registered hypotheses can be so large. To a first approximation, the count of distinct hypotheses in a study is the product of the populations or subgroups it examines, its interventions, treatment arms, outcomes, the contrasts drawn among the arms, and any pre-registered heterogeneity dimensions. Because these choices enter multiplicatively rather than additively, even moderately sized experiments can generate long lists of tests, which is reflected in the skewed distribution noted above (a median of 9 hypotheses per study but a mean of nearly 16). Although this combinatorial logic may seem trivial, and one might assume it is common knowledge, it is in fact not well understood within the field: among the experts surveyed for this project (Section 2.3), the median forecast of the number of hypotheses in a typical study was only 4, less than half of the 9 observed in the data, and 85% of forecasts fell below the observed median. (See Appendix B for more details.)

2.2 Experimental Design

An RCT was then conducted to investigate the primary constraints that prevent researchers from reporting all their pre-registered hypotheses. The study specifically examined three issues that may constrain complete reporting. The first is a lack of awareness or information regarding the scientific value of reporting all pre-registered hypotheses. The second is a lack of engagement from other scholars regarding their findings, which when combined with research time constraints may lead to less reporting. The third is the lack of research resources (i.e., funding, time) needed to either generate the results or to report them in project outputs and documentation.

The experiment was designed to test these constraints in a nested fashion in which each treatment arm builds on the last. Specifically, the RCT consisted of three treatment arms plus a control group.

In the **Awareness** arm, an email was sent to the corresponding author (as recorded in the AEA Registry) with a message on the importance of reporting all results of pre-registered studies (see Appendix, Exhibit A2). The message encouraged them to upload a report with results to the AEA Registry and suggested that they either (i) upload a paper in their preferred format or (ii) use a *Results Report* template provided by the study team (described below). The email included links to the empty Results Report template and an example of a pre-filled Results Report from another study.

In the **Engagement** arm, the same email message from the Awareness arm was sent, along with a personalized pre-filled *Results Report*. This email informed the researchers that the main hypotheses of their study had been encoded (by our team) in a standardized format and that the results corresponding most closely to these pre-registered hypotheses had also been recorded when they were publicly available. All this information was presented to the researchers in the pre-filled Results Report, and sharing the encoded

information with authors is what we think of as community engagement with their research. (Those in this treatment arm also received the same pre-filled Results Report example from another study, exactly as in the Awareness arm.)

The **Resources** arm builds on the Engagement arm. In addition to those interventions, study authors in this arm were offered up to 30 hours of research assistant (RA) support based at U.C. Berkeley, or up to \$1,500 USD to cover local RA time at their own institution. This support was framed as an offer to help the author carry out statistical analysis or review study documentation to obtain estimates for any missing results and complete or correct the pre-filled Results Report. Authors in this treatment arm had the choice to opt into the RA support, in which case they could select among the two RA options (i.e., at Berkeley or funds sent to their institution).

In the **Control** arm, there was zero contact with study authors in most cases and minimal contact for some. The authors of these studies were only contacted when the study team needed to request their private PAPs before the interventions began (which occurred in 14% of the control arm registrations).¹¹

A key innovation of this study is the **Results Report (RR)** noted above, which is a standardized document designed as a report card of sorts for a study's pre-registered hypotheses. Each page of the Results Report corresponds to one of the main hypotheses. The RR captures essential fields such as the hypothesis statement, results (coefficient estimate and standard error or p-value), and their location (in an article), reasons for missing results (if applicable), and any associated heterogeneity estimates. Every RR also contains links to a pre-filled example¹² and instructions for optionally posting these results to the AEA Registry¹³ (Appendix, Exhibit A18).

As described above, two distinct types of RRs were sent as part of the intervention. In the Awareness arm, participants received an *empty RR* attached to a generic cover letter. The document provided 20 blank pages for reporting hypotheses, with a link to request additional pages if necessary (Appendix, Exhibit A8). In the Engagement and Resources arms, participants also received a *personalized RR* beginning with a custom cover letter that specified the study title and the exact number of pre-registered hypotheses that the study team had encoded (Appendix, Exhibit A10) following the procedure described above. Following the cover letter, the document included a dedicated page for every main hypothesis encoded from the pre-registration. Each page contained the pre-registered hypothesis statement, along with a link to documentation describing the construction of the statement, including encoded populations, outcomes, interventions, arms, hypotheses, and any necessary judgment calls (Appendix, Exhibit A17). Hypotheses were grouped by whether results were found or not found, separated by a distinct page and differentiated by the title of each hypothesis page. When results for a registered hypothesis were found, the pre-filled results report presented the corresponding estimates and asked the study authors to confirm or correct them as necessary (example in Exhibit A11 in the Appendix). When results were not found for a given hypothesis (after detailed online search by the study team), the results report noted this and

¹¹Some authors note on the AEA Registry that they have a pre-analysis plan but it is not publicly available. This contact was necessary to generate comparable data for G_0 and we view it as unlikely to lead to major changes in researcher behavior given the ubiquity of such light-touch requests.

¹²Full pre-filled example here: <https://go.cega.org/RGPBFilledExample>

¹³And to demonstrate its feasibility, the Results Report for this study is produced in the Appendix A1

asked the authors to complete the report accordingly (example in Exhibit A14 in the Appendix).

Beyond the initial email contact and intervention described above, there was also a procedure for follow-up emails in cases where emails bounced or the corresponding author was unresponsive. All three intervention arms employed the same email follow-up procedure (note that the control group was not contacted). The initial email was sent to all publicly available e-mails for the main primary investigator, as listed in the AEA registration. In the event of no response after two weeks, a reminder was sent to the same e-mail address(es). If no response was received after three weeks (one week following the first reminder), a third and final email was sent, this time addressed to all researchers listed as primary investigators in the registration, based on an online search of available email addresses.

We followed a pre-specified approach to stratify the randomization along three dimensions: (1) pre-registrations with *versus* without a pre-analysis plan (PAP), (2) pre-registrations with a study population in a low- to middle-income country (LMIC) *versus* in a high-income country (HIC), which served as a proxy for studies in development economics, and (3) the identity of the encoder on the study team involved in the study.¹⁴

The randomization was also carried out in batches, as the encoding of pre-registrations proceeded over time during 2023-2025. Once the baseline encoding was completed for the first 140 studies, they were randomly assigned in equal proportions, to one of four arms (control and the three treatment arms). Subsequent batches were assigned a higher proportion to the three treatment arms (with only 5% in control and the other 95% assigned equally across the three treatment arms), to boost statistical power in the comparisons across the treatment arms.¹⁵

Most of the responses to our interventions reached us through a private channel rather than the public record.¹⁶

2.3 Collection of Expert Priors

To benchmark the findings against researchers' expectations, a forecast survey was conducted using the Social Science Prediction Platform (SSPP). A total of 68 experts participated, including researchers, graduate students, and practitioners in economics and related social science disciplines. Participants predicted the proportion of pre-registered hypotheses with publicly available results eight to nine years later, and the share of these publicly available results expected to be null. Respondents also stated how they expected these outcomes to vary across study characteristics, including the presence of a PAP,

¹⁴In the small number of cases in which a study was coded by more than one encoder, the study was randomly assigned to one of the encoders.

¹⁵This contingency was pre-specified in this study's PAP and was based on the study teams' prior that control group studies were unlikely to have meaningful updates in terms of study results between G_0 and G_1 , as 8 to 9 years had already passed after pre-registration. Initial examination of control group registration seemed to confirm that this was the case. A randomly chosen set of 35 studies was randomly archived for possible future encoding into the control group should substantial changes be observed in results reporting in the control group over time, see Figure 1

¹⁶Only 10.4% of the newly available results became publicly observable (in a paper, working paper, report, or registry-posted Results Report), and only 0.6% of the explanations came from studies with a public Results Report.

the study population (in a LMIC *versus* HIC), publication in any academic journal and publication in a top-five ranked economics journal.

Although the number of forecasters may appear to be relatively modest, prior work shows that structured prediction exercises with similar sample sizes can yield informative benchmarks for scientific expectations (DellaVigna et al., 2019). These expert forecasts serve as a reference point for interpreting the observed reporting patterns and assessing how closely the economics profession’s beliefs align with actual practices.

3 Empirical Strategy for the RCT

The empirical strategy for the RCT was pre-registered on the AEA Registry and the analysis largely follows these plans, although we note below where the analysis goes beyond the PAP.¹⁷ The discussion here begins by laying out the three main outcomes of interest in the RCT analysis. These capture the availability of results, the explanation for missing results, and the proportion of null results, and each is discussed in turn in what follows.

The *Fraction of hypotheses with available results* (**Available**) is computed at the study level as the share of pre-registered hypotheses (in the G_0 data) for which a numerical estimate is found in the follow-up (the G_1 data). This outcome was explicitly denoted as a primary outcome in the PAP. We attempted to define this variable as inclusively as possible: it counts a hypothesis as available if the result is found anywhere in the paper, appendix, or any other public record (including a results report emailed directly to the study team), regardless of the level of effort spent to find it, and also allows for some reasonable modification from the pre-registration provided that the encoders judged it to be broadly consistent with the originally pre-registered hypothesis. More stringent alternatives might only count results found with low effort (i.e., in the abstract) or reported in the main paper (rather than in the appendix), or that exactly match the original hypothesis, and these would lead to lower rates of available results.

The *Fraction of hypotheses with explanations for missing results* (**Explained**) is computed at the study-level as the share of hypotheses without an available result (i.e., *Available* = 0) that have a documented explanation for why it is missing.¹⁸ The explanations were provided by the authors of a particular study and include reasons such as the RCT as a whole was never carried out (or certain components were never carried out), endline data was never collected, and so on. These explanations were provided by the authors in free-text form (in response to emails from our study team) and were later grouped into one of seven main categories, namely: (1) no intervention occurred or a failed intervention; (2) the intervention occurred but no data was collected; (3) the intervention occurred but there is bad quality data; (4) the data was collected but analysis is yet to be carried out; (5) data was collected but statistical power is inadequate; (6) the authors disagree with our team’s assessment that a particular hypothesis was pre-registered; and (7) other explanations (in a small number of cases). In the main analysis,

¹⁷The pre-registration (AEARCTR-0013755) can be found at <https://www.socialscienceregistry.org/trials/13755>.

¹⁸This outcome was not denoted as a primary outcome in the PAP; it has proven useful in allowing us to better understand why results are not available in the G_1 data.

we count a missing hypothesis as explained only if it falls into the first two categories — no (or failed) intervention and no data collected — as these reflect circumstances most clearly outside of the researchers’ control; Table 7 examines how the estimated treatment effects change as the remaining categories are progressively included.

The *Fraction of hypotheses with null results* (**Null**) is computed as a study-level mean as the share of hypotheses for which the associated p -value is greater than 0.05 (the conventional level of statistical significance). When the p -value is not reported directly, an attempt is made to compute it on the basis of any available estimates and standard errors. When estimating the causal effects of the RCT on the fraction of null results, non-available hypotheses are included in the sample and assigned a value of zero, i.e., non-reported results are not considered null results. In some cases below, we also consider the fraction of null results conditional on a result being available.

With these primary outcomes in hand, the empirical strategy seeks to test three main hypotheses for each, closely following the RCT design described above. First, is there an effect of the *Awareness* treatment arm? Second, does the *Engagement* treatment arm have an additional effect (beyond that of the *Awareness* arm)? And third, is there an additional effect when authors are offered *Resources* (beyond any impact of *Engagement*)? As notation, in the rest of this section, the four arms are referred to as T_0 (Control), T_1 (Awareness), T_2 (Engagement), and T_3 (Resources). The PAP also specified that, if there is no significant additional effect of *Resources*, then the analysis would pool the *Engagement* and *Resources* treatments into one *Pooled Engagement* arm to improve statistical power for the comparison with the *Awareness* arm and versus control.

The analysis begins by considering an outcome Y (i.e., **Available**) that is observed both before and after the intervention. $t = 0$ indicates measurement in the pre-intervention period while $t = 1$ is post-intervention. So $Y_{i,t=0}$ is the outcome for study i before the interventions were carried out and $Y_{i,t=1}$ is the outcome post-intervention.

As laid out in the PAP, the main regression specification is difference-in-differences:

$$Y_{it} = \mu_i + \delta_1 I_{t=1} + \sum_{j=1}^J \tau_j (T_{ij} \times I_{t=1}) + \varepsilon_{it}, \quad t = 0, 1, \quad (1)$$

where T_{ij} is an indicator for study i being randomized into treatment arm j , $j \in \{0, 1, 2, 3\}$, i.e., the three treatments plus the control group, where effects are relative to the left-out control arm ($j = 0$). $I_{t=1}$ is an indicator for the $t = 1$ post-period. The specification contains study-level fixed effects (μ_i) and a time effect for the post-intervention period (captured in the δ_1 term). Unbiased treatment effect estimates can also be obtained using only the post-intervention cross-section of data:

$$Y_{i1} = \mu_0 + \sum_{j=1}^J \tau_j T_{ij} + \zeta_s + \varepsilon_{i1}, \quad (2)$$

where indicator variables for each of the randomization strata (ζ_s , LMIC, PAP and encoder indicator variables) are included.

For an indicator variable Z_i capturing a subgroup of studies – e.g., those with PAPs, or those with a study population in LMICs, dimensions pre-specified in the PAP – het-

erogeneity in treatment effects is estimated by extending equation (1) as follows:

$$Y_{it} = \mu_i + \delta_1 I_{t=1} + \delta_{1Z} (Z_i \times I_{t=1}) + \sum_{j=1}^J \tau_j (T_{ij} \times I_{t=1}) + \sum_{j=1}^J \gamma_j (Z_i \times T_{ij} \times I_{t=1}) + \varepsilon_{it}, \quad t = 0, 1. \quad (3)$$

Heterogeneous treatment effects can also be estimated using the post-intervention cross-sectional data in an analogous way.

The following hypotheses are tested based on the main regression specification (Equation 1). H1, H2, and H3 were pre-specified as the primary hypotheses.¹⁹ For each, the null hypothesis (H_0) is listed first and the alternative (H_A) next to make explicit if the tests are one- or two-sided, i.e., $H_A : \beta \neq 0$ is two-sided and $H_A : \beta > 0$ is a one-sided test, where the latter one-sided hypotheses were considered the primary tests in the PAP.

H1 There is no effect of the informational treatment together with the provision of an empty results report template (T_1):

$$H1_0 : \tau_1 = 0, \quad H1_A : \tau_1 > 0. \quad (4)$$

H2 There is no incremental effect of provision of the pre-filled results report (T_2) relative to T_1 .

$$H2_0 : \tau_2 - \tau_1 = 0, \quad H2_A : \tau_2 - \tau_1 > 0. \quad (5)$$

H3 There is no incremental effect of the offer of RA support (T_3) relative to T_2 .

$$H3_0 : \tau_3 - \tau_2 = 0, \quad H3_A : \tau_3 - \tau_2 > 0. \quad (6)$$

4 Empirical Results

The analysis begins with a discussion of the descriptive results in section 4.1, before turning to the experimental results (Section 4.2) and the analyses of heterogeneity and spillovers that follow.

4.1 The Size of the File Drawer in Economics

Figure 2 provides an overview of the level of under-reporting of results from hypotheses pre-registered on the AEA Registry during 2015-2017. The full sample of pre-registrations we consider is illustrated in Panel A, consisting of 317 studies and 5,048 hypotheses. Based on the follow-up data collection activities (G_1), 19.2% of these studies have no paper publicly available in any form — published paper, working paper, report, etc. — 8–9 years after pre-registration (Panel B).

¹⁹Note that the hypotheses H2 and H3 were labeled “H4” and “H5” in the PAP.

[FIGURE 2 HERE]

Focusing on the existence of a paper alone, however, may overestimate reporting at the hypothesis level. The next step of the analysis thus distinguishes between studies with a paper where results are available for all pre-specified hypotheses, only some hypotheses, or none (Panel C). Among the 80.8% of studies with a paper available, 75% do not report all pre-registered hypotheses. Among all studies, only 20.2% report results for all of their pre-registered hypotheses, which is 25% of the 80.8% of studies with a paper available. Most studies report at least some pre-registered hypotheses: 68.5% of all studies have at least one pre-registered result available (and 80.8% have a paper available). This leaves 12.3% of the full sample with a paper but no reported pre-specified results. This implies that no pre-registered hypotheses are available for nearly a third of all studies (specifically, 31.5%, taking the 19.2% with no available paper and the 12.3% with no reported results despite having a paper).

Panel D presents the fraction of pre-specified hypotheses with available results (**Available**) for each pre-registration in the sample. Averaged across all studies, only 42% of pre-registered hypotheses have available results 8–9 years after registration, or conversely that the size of the “file drawer” corresponds to 58% of hypotheses. We view the construction of this figure as one of the novel contributions of this study.

Figure 3 presents the distribution of the share of *Null* results conditional on results being available for the hypothesis (i.e., hypotheses with no results available are excluded). On average, 64.5% of available results are null (at the conventional significance level of $p > 0.05$). This evidence challenges the widespread view that relatively few available research results in economics are null. In fact, for almost 30% of these studies, nearly all reported results are null (90 to 100% of results). This is a second noteworthy finding of the descriptive analysis.

[FIGURE 3 HERE]

The analysis next examines variation in the fraction of available results and the fraction of null results across several dimensions (Table 2).

[TABLE 2 HERE]

A first observation is the somewhat mechanical result that the fraction of hypotheses with available results increases when the sample is restricted to pre-registrations with some output available (“Pre-registrations with paper”), rising to 52%. Note, however, nearly half of all hypotheses remain unreported even in this case, implying that missing results cannot be explained solely by studies that were never carried out.

Second, working papers and published papers not in the top-5 economics journals show broadly similar fractions of available results (at 51.6% and 55.1%, respectively) as well as null results (conditional on availability) at 66.7% and 65.5%, respectively. In stark contrast, those articles published in the top 5 journals exhibit quite different patterns, with far lower rates of availability (41.3%) and fewer null results conditional on availability (51.3%). This pattern is consistent with a meaningful degree of publication bias for studies in these top journals, although note that there are many potential confounding factors. The difference between top 5 and non-top 5 studies in terms of the

share of null results is also only marginally statistically significant ($p\text{-value} = 0.034$), so taken together this offers only suggestive evidence of publication bias in the top economics journals.

Third, increasing the stringency of what counts as an available result also has sizable effects on both the fraction available and the fraction of null results. For instance, only counting results that appear in the main body of the paper reduces the fraction available from 42% to 34.8% and the fraction null from 64.5% to 61.1%. The encoders also classified results in terms of how closely they aligned with the original pre-registration, and only counting hypotheses that were judged to have not been modified reduces availability much further to 28.6%, while slightly increasing the fraction null (to 66%). Imposing both of these restrictions and additionally requiring the result to be “not hard to find” (a subjective assessment by the encoders) reduces the fraction available to 19.7%. Thus the most stringent definition implies that 80% of pre-registered hypotheses are not readily available.

Fourth, investigation of primary pre-registered heterogeneity hypotheses reveals several insights. When the heterogeneity hypotheses are removed, the *Available* rises slightly, while the fraction null (*Null* | *Available*) is essentially unchanged. Among the subset of $N = 39$ studies with pre-registered primary heterogeneity hypotheses, the share of available results is much lower overall (at 31.4%), and this is driven entirely by the heterogeneity hypotheses themselves, which have an availability rate of only 22.3%. However, there are no meaningful differences in the fraction of null results even in this case.

Finally, when examining other study-level characteristics, studies with a pre-analysis plan (PAP, which recall tend to have more detailed hypotheses than many pre-registered studies without a PAP) have essentially the same or only a slightly higher fraction of results available (at 43.9%) than studies without a PAP. In contrast, studies conducted in low- and middle-income countries (LMICs) – which are nearly two thirds of the overall sample of pre-registrations – and studies with many (more than 10) pre-registered hypotheses tend to have fewer available results, at 37.8% and 31.8%, respectively. The latter finding is consistent with the view that authors who list out a large number of tests *ex ante* spend less effort on average making sure each one gets reported, relative to studies that only pre-specify a small number of primary tests, suggesting a potential quality-quantity trade-off in writing pre-registrations.

Figure 4 compares these findings to the expectations of experts surveyed as part of this project through the SSPP (as described above). The upper two panels show their predictions for the fraction of available results and the fraction of available null findings. Panel A shows that the experts believe that there is even less reporting on average than we document in the study: the experts believe that the average fraction of available hypotheses is 27%, compared to the 42% estimated in the data. Only a small minority of the experts expected more results to be available than is actually the case. The difference between these two rates is statistically significant at 95% confidence.

[FIGURE 4 HERE]

We observe that experts are also pessimistic about the availability of null results, with an even larger gap between priors and the data. They expect only 37% of available

findings will be null results — reflecting the widespread view that few null results ever get published — while we show in the data that the fraction is almost double at 64.5%. The difference between these rates is again significant ($p\text{-value} = < 0.001$).

The lower two panels then show how experts expected these outcomes to vary with different study characteristics. We observe that experts expect the registration of a PAP to have a clear positive effect on both the fraction of available results (Panel C) and the fraction of available null findings (Panel D), whereas the data shows no difference for *Available*, and a small positive effect on the fraction null that is not statistically different from zero. There are no significant differences in available results or null results for articles published in non-top 5 journals (relative to working papers). Experts did not expect to see major differences on average in the reporting of results for papers published in either non top-5 or in top-5 academic journals, but as we showed above there is far lower reporting of results in top-5 journals (and this effect is significant, Panel C). The experts expected all published papers, especially the top-5 economics journals, to have fewer available null results, and this is the case for top-5 journals (with a marginally significant effect), but not for journals outside of the top-5.

In general, experts did not have strong directional expectations regarding how reporting of results would differ for studies carried out in LMICs (i.e., development economics) but as noted above there are significantly fewer results available for these studies, but no difference for null results. Finally, we present the results for studies with many hypotheses (more than 10), a dimension that we explore after recording our own PAP, showing that there tends to be far less reporting of findings in this case (and suggestive evidence of more reporting of null results).

4.2 Experimental Results

Table 3 presents summary statistics detailing author engagement with the three treatment interventions. Across all three arms, engagement was high: 89.1% of the studies had at least one author click on any link within the email(s) sent (Column 2). Results Report-specific click rates (Column 3) were also substantial across all arms, averaging 88.3%. Furthermore, about half of the recipients of the pre-filled results reports (Engagement and Resources arms) clicked on the encoding details of their own study (Column 4: 52.3% and 47.7%, respectively). In terms of email responses (Columns 5–7), the Engagement and Resources treatments led to a higher volume and more informative responses than the Awareness treatment. Specifically, around half of the studies in the Engagement (60.2%) and Resources (50%) arms provided an informative response (Column 6), compared to 28.9% in the Awareness arm. In all, the intervention led to meaningful engagement, with nearly nine out of ten author teams engaging in at least some way with the materials provided in the email and nearly two thirds directly responding via email.

[TABLE 3 HERE]

Given that this was a meaningful intervention, and authors did not just ignore the emails, we next estimate treatment effects. Figure 5 shows the coefficient estimates for all treatment arms using the preferred specification (Equation 1). The *Awareness* arm did not produce any increase in the number of available estimates, with an effect very close to zero. The *Engagement* and *Resources* treatments, however, yielded positive effects remarkably similar to each other and close to 6%, both of which are statistically significant at over 95% confidence relative to the control group. Given that the size of the documented file drawer was 58% percentage points, these effects represent a reduction in the size of file drawer of around 11%, i.e., (6.2 percentage points) / (58% percentage points).

[FIGURE 5 HERE]

Table 4 presents the main regression results for the outcomes *Available* (Columns 1–4), *Explained* (Columns 5–6), and *Fraction Missing* (Columns 7–8), starting with *Available*. Column (1) reports difference-in-differences estimates. Consistent with the graphical evidence, the Awareness arm did not produce any increase in available estimates. The Engagement and Resources treatments showed similar and positive effects close to 6 percentage points, both of which are significant. The pre-specified test for the incremental difference between the Engagement and Awareness arms rejects equality (row H_2 , p-value = 0.001), but the test for the incremental effect of Resources beyond Engagement (row H_3) is not statistically significant. We thus pool these two treatments into a single arm in Column (2), following the PAP. This pooled treatment yields a preferred estimate of a 6.2 percentage point increase (significant at over 99% confidence). Columns (3) and (4) report the corresponding cross-sectional estimates (Equation 2), which replace the study fixed effects with the pre-specified controls and the baseline level of *Available*; the estimates are nearly identical to the difference-in-differences results.

[TABLE 4 HERE]

Columns (5) and (6) present the cross-sectional estimates for the outcome *Explained*. Since this was only available in the follow-up G_1 data, we rely on cross-sectional estimation (Equation 2). The control group mean for this outcome was 0, meaning no explanations were found for hypotheses with missing results in the control group. The Awareness arm resulted in a significant 5.2 percentage point increase ($p = 0.012$). The Engagement (12.3 percentage points) and Resources (8.3 percentage points) arms yielded even larger increases, though with no statistically significant difference between them (row H_3). Column (6) thus again reports the pooled estimate for Engagement and Resources, yielding an effect estimate of 10.3 percentage points (significant at over 99% confidence). This pooled effect represents a sizable 18% reduction of the file drawer, i.e., (10.3 percentage points) / (58% percentage points).

Finally, Columns (7) and (8) report the estimates for the outcome *Fraction Missing*, defined as the share of pre-registered hypotheses that are neither available nor explained; its complement, $1 - \text{Fraction Missing}$, is the combined share of hypotheses that are either available or explained. These outcomes are estimated using the cross-sectional specification (Equation 2), under which the three outcomes form an exact linear partition, $\text{Available} + \text{Explained} + \text{Fraction Missing} = 1$. The Column (7)–(8) coefficients are therefore simply the negative of the sum of the corresponding cross-sectional *Available* (Columns 3–4) and *Explained* (Columns 5–6) effects, and add no information beyond them. We report *Explained* and *Fraction Missing* only under the cross-sectional specification because *Explained* is observed only in the post-intervention (G_1) period, which precludes a difference-in-differences estimate of the combined outcome. Relative to a control-group mean of 0.64, the pooled Engagement and Resources treatment reduces the fraction of missing results by 16.4 percentage points (Column 8, significant at over 99% confidence), the sum of its 6.1 percentage point effect on *Available* (Column 4) and its 10.3 percentage point effect on *Explained* (Column 6). Of this available portion, only 2.5 percentage points reflect estimates that were genuinely new at baseline (see the decomposition in Table 5); the remainder existed but was buried or had already been observed. The Awareness arm also significantly reduces the fraction missing, by 5.7 percentage points, even though it had essentially no effect on *Available* (an increase of just 0.4 percentage points); its entire effect on this outcome therefore operates through explanations. Because a hypothesis leaves the missing category when authors supply either a result or an explanation, a lower *Fraction Missing* indicates that the status of a hypothesis is now known, not necessarily that its result has been produced.

As in the other columns, the incremental effect of *Resources* beyond *Engagement* is not statistically significant (row H_3 , $p\text{-value} = 0.868$). Note that, unlike *Available*, neither *Explained* nor *Fraction Missing* was pre-registered, so these two outcomes should be viewed as exploratory.

Estimates for the second pre-registered outcome, *Null* (the fraction of hypotheses with null results reported), are presented in Panel A of Table 6. Corresponding to the lack of effect on overall availability, the Awareness arm produced no increase in null results reporting. Treatment effects for the Engagement and Resources arms (Panel A, col. 1), and the Pooled arm (col. 2) are all positive and statistically significant, at around 4.2 percentage points. A comparison of the pooled estimate for *Available* (col. 2 of

Table 4) with the pooled estimate for *Null* suggests that most of the results that are newly reported due to the interventions are null findings.

We adjust for multiple hypothesis testing following our pre-analysis plan (5.4). A joint Wald test from the SUR specification rejects the null that all six treatment effects (three arms $\times$ two outcomes) are zero ($\chi^2(6) = 20.52$, $p = 0.002$).

After Romano-Wolf FWER correction, the engagement effect (H2; H4 in the PAP) remains significant for both availability ($p = 0.001$) and null reporting ($p = 0.003$), while the awareness effect (H1: $p = 0.164$, $p = 0.208$) and the resources effect (H3; H5 in the PAP: $p = 0.746$, $p = 0.826$) do not survive adjustment.

As a robustness check, we extend the MHT framework to include the explained outcome alongside the two pre-registered outcomes. The joint Wald test across all nine treatment effects ($\chi^2(9) = 50.4$, $p = <0.001$) rejects the null.

Romano-Wolf adjusted p -values for the nine hypotheses are: H1 (awareness): availability $p = 0.228$, null reporting $p = 0.277$, explained $p = 0.071$; H2 (engagement – information): availability $p = 0.006$, null reporting $p = 0.015$, explained $p = 0.104$; H3 (resources – engagement): availability $p = 0.915$, null reporting $p = 0.972$, explained $p = 0.972$.

Newly available hypotheses can be subject to different classifications, similar to the Explained outcome. Table 5 presents these different types and reports the pooled treatment estimates by cumulatively increasing the categories from top to bottom. Newly available hypotheses fall into three main categories: those not available at baseline; those available but not possible to find without further information from the authors (labeled as buried); and those available as sent by the authors in response to the interventions, but already observed at baseline and deemed inconsistent by the research team.²⁰ The distinction matters for interpreting the headline effect. Only 2.5% of the pooled treatment effect (of Engagement and Resources) corresponds to estimates that were genuinely new, absent anywhere at baseline; this is the share the interventions truly recovered. Adding results that existed but were buried raises the effect to 4.8%, and adding results that had already been observed but were deemed inconsistent yields the final 6.2% reported in Table 4. Thus, of the 6.2 percentage point headline increase in availability, less than half reflects results that did not already exist in some form. All three estimates remain statistically significant.

[TABLE 5 HERE]

To investigate in greater depth whether the newly available results contain a higher fraction of null findings than other reported findings, Panel B of Table 6 presents the average fraction null at the study level, before and after the interventions, by including all hypotheses (unconditional) and restricting only to hypotheses with available results (conditional). The regression estimates discussed above (Panel A) captured the difference in the fraction null (unconditionally) over time, where hypotheses with unavailable results are included as zeros. When examining the composition of null results among hypotheses with available results (conditional null), the pre- and post-intervention conditional fractions are actually quite similar across all treatment arms. Furthermore, fo-

²⁰Due to a design constraint in the results report, it was decided that in case of inconsistent modifications, the pre-filled report sent to the authors would display the corresponding hypothesis as not found.

cusing only on the hypotheses with newly available results and comparing their fraction null per study to the conditional baseline rate reveals only a slightly higher share of null results among the newly available results, at 68.4% (col. 5) versus 64.5% (col. 2). While this is a non-trivial difference, it is not statistically significant at traditional confidence levels (col. 6, $p\text{-value} = 0.563$ overall) and they vary directionally between the Engagement and Resources arms. Hence, and perhaps surprisingly given the view that null results are often suppressed in the empirical economics literature, the share of null results among newly available estimates does not appear meaningfully different from the hypotheses available before the intervention.

[TABLE 6 HERE]

We next examine the explanations by authors regarding why results were not available in more detail. These explanations were grouped into the seven categories introduced above; the six substantive categories — ranging from reasons most likely outside of the researchers’ control (e.g., “Failed or no intervention”) to those that appear under their control (e.g., data is available, but “low power” or yet “to do analysis”) — are analyzed in Table 7, while a small residual of other (“not pre-registered”) cases is excluded there, as noted in the table. Table 7 shows the distribution of explained hypotheses, and reports the pooled treatment coefficient (Engagement or Resources) when progressively including additional explanation categories.

[TABLE 7 HERE]

We first focus on explanations for missing results that reveal that study interventions — either the entire RCT or specific arms — were not successfully implemented. These could happen for a variety of reasons, including failed partnerships with government or NGOs that was supposed to carry out the intervention(s), or implementation difficulties within a program. These are quite common causes for why results are not available, accounting for 18.2% of all explanations provided for a missing hypothesis (and being relevant for 31.2% of all studies). The effect of the pooled Engagement or Resources arms, relative to the control group (where there is essentially zero reporting of explanations), is 5.5 percentage point (SE 1.8 percentage points, $p\text{-value} < 0.01$). These arms lead to a sizable increase in knowledge about which results are not available because the intervention never occurred or failed. Such explanations do not reduce publication bias, since no new estimate is produced, but they are valuable in their own right: a public record that a planned intervention was never carried out or failed helps other researchers avoid duplicating abandoned work, reducing research waste.

A second leading cause for results not being available is the lack of data to test various pre-specified hypotheses, even in cases where the intervention(s) did occur as planned. Authors provided many reasons for why no endline data was available, including the project running out of funding to collect the data, facing insurmountable logistical constraints in the field, as well as cases in which no one took up the intervention (even if it was implemented) making it impossible to test impacts. This was the main explanation for unavailable results for 31.2% of hypotheses, so taken together these first two categories — no intervention or no data — account for nearly half of all

explanations provided for missing results (49.5%). The pooled effect of the two treatments (Engagement and Resources) on reporting across these first two categories is 10.3 percentage points (SE 2.4, p -value <0.01). Recall that the share of available results in the data is 42% of pre-specified hypotheses. If we take into account the fact that either interventions or data collection did not occur for 10% of all hypotheses, the share of results reported among those that could potentially have been reported is somewhat higher at 47%, reducing the size of the file drawer from 58% to 53%.

These first two explanations for unavailable results are relatively uncontroversial in the sense that it seems likely that authors are simply unable to report results from hypotheses (and studies) where key interventions and data collection simply did not occur. The other half of explanations appear to be more of a gray area, in the sense that some additional reporting seems feasible. For instance, in 6% of hypotheses, authors reported that interventions occurred and data was available but that the data collected was of “bad quality”, for various reasons. It might still be possible for results to be reported in this case, though perhaps data collection difficulties, i.e., low quality enumerators, data collection errors in the field, etc. would make the results unreliable. An immediate concern, though, is that researchers consider the data “poor quality” because they did not obtain the expected results, including in terms of statistical significance. Although the current research team cannot adjudicate between these explanations (since we do not have access to this unreported data), failing to report results because of the nature of the empirical findings would constitute selective reporting.

A closely related case is one where there is data but the authors claim they did not report results because the findings have “low power” statistically, and these account for 19.4% of all explanations for unavailable hypotheses. Once again, it is possible this is due to truly poor quality data or research design flaws but the concern arises that “low power” is another way of saying that the authors are choosing not to report results that are not statistically significant at traditional confidence levels. These two data quality related cases account for 25.4% of unavailable hypotheses in all, a meaningful share.

The final large category of explanations — accounting for 20.4% of unavailable hypotheses — are cases in which the data was collected (and interventions carried out) but the authors explain that they are yet to do the analysis. Here the authors are explaining that they could do the analysis but have not yet done so, so these are cases where analysis could be carried out (unlike the cases where interventions or data collection never occurred). A final category (4.7% of explanations for unavailable hypotheses) is when our team of encoders concluded that pre-specified results were not reported but the original study authors responded that they had been reported. Many of these cases occurred when our team viewed modifications in outcome measures or tests as being sufficiently meaningful that they no longer corresponded to the pre-specified tests.

The table presents the results for the impact of the pooled Engagement and Resources treatment on the share of explanations for unavailable results including this set of more controversial “gray area” cases (beyond simply no intervention or no data collected). The overall effect across all categories is an increase of 15.2 percentage points in the share of results with explanations (SE 2.8 percentage points, p -value <0.01 , Table 7). Regardless of whether the explanations are viewed as valid reasons for not reporting results, a positive interpretation is that at least the interventions increased understanding of why results were not available (and could lead to productive discussions with

study authors about further analyses). With the loosest definition including all categories, these interventions reduce the size of the file drawer from 58% to approximately 48%.

It is also useful to look at the combined effect of the pooled Engagement and Resources interventions on both the availability of results and the explanations for missing results. Counting only actual results, the share of hypotheses with publicly available findings rises modestly, by 6.2 percentage points from the original 42%. The combined figure exceeds half only once explanations for why results are absent (narrowly defined as cases where the intervention or data collection did not occur) are also counted; these explanations establish that a result is missing and why, not that the result has been produced. Employing the less conservative definition including “gray area” cases would account for a larger share still, on the same caveat.

4.3 Heterogeneous Treatment Effects

Table 8 presents the regression estimates for the heterogeneity analysis across the pre-registered dimensions of having a pre-analysis plan and being implemented in an LMIC. Treatment effects are reported for the studies that *do not* belong to the heterogeneity group of interest (e.g., studies that do not have a PAP), and for those studies that do belong to the group, as well as the p -value on the null hypothesis of equal effects for both groups. These are presented for each of the three main treatment arms (Awareness, Engagement and Resources) as well as for the additional effects of Engagement (beyond Awareness) and Resources (beyond Engagement), mirroring hypotheses H2 and H3 in the main analysis.

The bottom line is that there is little evidence of heterogeneous treatment effects across these two pre-specified dimensions: across the 30 tests of heterogeneity presented in the table (represented in the rows with “p-value for Yes=No”), only 2 are statistically significant at the traditional 95% confidence level — roughly the number of false positives that would be expected by chance. The 2 significant cells fall in different treatment arms, outcomes, and subgroup dimensions: the Resources effect on result availability is larger for studies with a PAP (a 12.4 percentage point effect, versus 0.6 for studies without one; $p = 0.015$), while the Awareness effect on explanations for missing results is concentrated among studies in LMICs (6.9 percentage points, versus 0 otherwise; $p = 0.032$). Because these isolated rejections are scattered across unrelated arms and outcomes rather than forming a coherent pattern, we read them as consistent with chance rather than as robust heterogeneity (a point the conclusion revisits).

Beyond this null bottom line, two features of the table are worth noting. First, the headline treatment effect is stable across subgroups: the Engagement arm raised result availability by 6.6 percentage points among studies without a PAP and 7.3 among those with one ($p = 0.874$), and by 7.8 percentage points outside LMICs and 6.4 within them ($p = 0.737$), with no significant difference on either dimension. The main finding that hands-on engagement recovers results is therefore not driven by any single subgroup. Second, the additional-effect rows confirm the main-text contrasts within subgroups: the increment from Awareness to Engagement (mirroring H2) remains significant for availability in every subgroup (all $p \leq 0.047$), whereas no Resources-over-Engagement increment (mirroring H3) is significant in any subgroup or outcome (the smallest such p

is 0.094). The limited heterogeneity that does appear should be read with extra caution for the *Explained* outcome, which was not pre-specified and is estimated with the cross-sectional rather than the difference-in-differences specification.

[TABLE 8 HERE]

4.4 Spillover Effects of Treatments on Researcher Behavior

A potential concern in interpreting the main treatment effects is that the interventions may have generated spillovers across studies through co-author networks. While we excluded studies sharing a primary author from our sample, pre-registrations may still be connected through co-author networks. A co-author on a treated study may have communicated relevant information or reporting norms to co-authors involved in other registered studies, potentially complementing our treatments through peer networks. Conversely, a study assigned to the control arm may share co-authors with treated studies, thereby contaminating the counterfactual and causing the main estimates to understate the true direct effects.

To assess the extent of this concern, we first characterize the co-author overlap structure among the 317 studies in the sample. Of these, 182 (57.4%) share no co-authors with any other registered study and are therefore not subject to network-based spillovers.²¹ The remaining 135 connected studies are sparsely linked: 85.2% share co-authors with only one or two other studies, and the most connected study shares co-authors with 5 others. Among connected studies, average total network exposure is $\bar{P}_i = 1.9$ (median = 2), while average exposure to treatment-arm studies is $\bar{S}_i = 1.57$ (median = 1). Appendix Figure A21 displays the distributions of S_i and P_i across treatment arms. Arm-specific panels confirm that T_k -exposure is highly concentrated at zero for all three treatment arms, consistent with the sparse network structure described above. The bottom row shows that the spillover index S_i and network size P_i are similarly left-skewed, with the large majority of studies having no or minimal co-author overlap with the rest of the sample.

To test for spillovers, we estimate two pre-registered specifications by expanding the main difference-in-differences equation with interaction terms. Addendum Equation 3 describes spillovers via β_0 , the spillover effect of any treatment exposure onto control-arm studies, and θ , which captures complementarity ($\theta > 0$) or substitution ($\theta < 0$) between own treatment assignment and co-author exposure. Addendum Equation 4 estimates arm-specific exposure interactions and is used to test the omnibus null of no spillovers. Standard errors are clustered at the study level throughout.²²

Table A4 in the appendix presents the main spillover results across both outcomes. The results provide no meaningful evidence of spillover effects through the co-author network. The omnibus Wald test of joint significance across all arm-specific exposure

²¹The realized network is sparser than the network simulated when designing the spillover analysis in the addendum to the PAP. We proceed with the pre-registered estimation strategy regardless.

²²Addendum Equations 3 and 4 are pre-specified in the addendum to the PAP; we label the spillover hypotheses S-H1–S-H4 to distinguish them from the main-text hypotheses. S-H1 is tested using Addendum Equation 4; S-H2, S-H3, and S-H4 are tested using Addendum Equation 3. We consider S-H1, S-H2, and S-H4 as primary hypotheses; S-H3 is secondary, as its result follows closely from S-H2 and S-H4 taken together.

interactions (S-H1, Addendum Eq. 4) fails to reject the null for both the fraction of results available ($p = 0.703$) and the fraction of null results reported ($p = 0.781$). Because there is no change in the control group over time, the spillover-onto-control parameter $\hat{\beta}_0$ is zero (S-H2, $p = 1$). The pre-registered test of no net spillover effect on treated studies (S-H3: $\beta_0 + \theta = 0$) also fails to reject ($p = 0.33$ for availability; $p = 0.296$ for null results). The complementarity parameter $\hat{\theta}$ is positive but statistically indistinguishable from zero (S-H4, one-sided: $p = 0.165$; $p = 0.148$), offering no evidence of peer-network amplification of treatment effects. Across both equations, the direct treatment effects for Engagement and Resources remain positive and significant, consistent with the main estimates in Section 4.2.

5 Conclusion

This study is among the first to provide systematic evidence on the availability of results at the hypothesis level for pre-registered studies in economics. While a limited body of prior research has quantified the fraction of studies without any public output, and found that this share is typically around 20% (Franco et al., 2014; Leight et al., 2025), the present analysis goes much further by evaluating whether pre-specified hypotheses can be linked to publicly available results. This more fine-grained approach enables us to create a richer and more direct measure of the file-drawer problem in economics.

In the study’s first main finding, we show that the majority of pre-registered hypotheses do not have available results nearly a decade after pre-registration: on average, 58% of hypotheses are missing. This approach also highlights that focusing solely on whether any paper exists—as most previous studies have done—may substantially underestimate the extent of under-reporting, by roughly three-fold, i.e., roughly 19.2% of projects have no available paper but nearly 58% of hypotheses do not have available results. This contribution complements work in other disciplines that has also provided strong evidence of under-reporting of results (Thibault and TARG Meta-Research Group, 2023; van den Akker et al., 2023), although much of that work relied on different and typically less detailed information than the present study.

Another central finding is that the under-reporting of null results does not appear to be the primary driver of missing evidence. On average, 64.5% of available results found by our team are null (non-significant) findings, and for nearly 30% of studies all or almost all reported results are null. We view this as a surprising finding, which is backed up by the expert forecasts that were elicited: experts predicted far lower rates of available results than documented. They predict 73% of results would be missing versus the 58% we show, and they expected there to be far fewer null results among reported findings (at 37%) than are actually observed (64.5%). The gap between these figures is one measure of how much one can learn from this study’s data.

We then conducted a randomized controlled trial (RCT) to provide causal evidence on interventions that might increase the reporting of results. Three treatment arms were designed to shed light on the main constraints behind reporting, namely, Awareness, Engagement, and Resources. The Awareness arm had no meaningful effect on the availability of results, but showed a statistically significant impact on the proportion of explained missing results, with an increase of 5.2 percentage points. Pooling the two in-

interventions that had larger impacts (namely, the Engagement arm and the Resources arm), these treatments increase result availability by 6.2 percentage points, or by 11% on a base of 58% missing results. There was no significant difference between the Engagement and Resources arms in terms of either results availability or the proportion of missing results with explanations, suggesting that a lack of financial resources to carry out pre-specified analysis is not a leading barrier to reporting. Pooling the Engagement and Resources arms, authors also provided explanations for a further 10.3 percentage points of the missing hypotheses (18% of the 58% file drawer), most often that the planned intervention or data collection never took place. These explanations account for why a result does not exist; they are not themselves recovered evidence, but they contribute in a different way: by creating a public record of which planned analyses were abandoned or never carried out, they can reduce duplicated effort and research waste even though they do not directly address publication bias. Taken together, the interventions reduce the share of hypotheses that are neither available nor explained by 16.4 percentage points (from 58% to 42%), as illustrated in Figure 6. This combined figure mixes two different things: roughly 6.2 percentage points of newly available results (of which only 2.5 percentage points were genuinely new, with no trace at baseline) and roughly 10.3 percentage points of explanations for why results are absent. Even after the most intensive arm, 42% of pre-registered hypotheses remain entirely unaccounted for, and the hands-on engagement required to achieve even this was substantial: neither awareness nor additional resources substituted for it.

[FIGURE 6 HERE]

There has been an active debate in economics about the value of a detailed pre-analysis plan (PAP) above and beyond a simple entry on the AEA RCT Registry (Banerjee et al., 2020). We find little systematic heterogeneity in the response to the interventions along this dimension: the one suggestive pattern is that the Resources arm raised the availability of results more among studies that had a PAP than among those that did not. This is an isolated result among many subgroup tests, however, and may partly reflect the selection of authors into writing a PAP, so the findings on heterogeneity by having a PAP should be interpreted with caution, especially given the limited sample size in subgroups of studies. For the most part, there are limited differences in impacts for research in development economics (proxied by those studies carried out in low and middle income countries, LMICs) versus other types of studies.

The data collection and encoding carried out in this project have created a novel dataset of pre-registrations and results in economics. One lesson from this study is that doing so is not cheap in terms of time and human resources: as noted above, encoding each paper conservatively took at least 11 hours of research staff time on average. In the future, it would be useful to explore an encoding approach that harnesses large language models (LLMs) or other artificial intelligence tools to reduce these costs (perhaps along the lines of <https://regcheck.app>), if it can be done without meaningfully sacrificing accuracy. This could make the approach taken in this study’s RCT a far more cost effective way to boost the reporting of scientific results at scale.

We also view the reporting standardization approach developed in this study—including both the schema used to capture hypotheses and the *Results Reports* templates

used to recover missing results—as steps forward in economics that might be useful in other contexts. In other disciplines, including medicine and public health, standardizing how results are reported is already well established and is viewed as an essential way to connect scientific evidence with policy and practice. The broad interest in the standardization of results in economics is indicated by the fact that there have recently been multiple attempts and calls to better harmonize evidence (Snilstveit et al., 2016; Vivaldi, 2020; Hendren and Sprung-Keyser, 2020; Edlin and Love, 2022). However, there is currently limited consensus regarding the best practices around the development, refinement, and use of reporting standards in economics. Various components presented in this study—namely, the metadata schema, the *Results Reports* templates, the data generated on thousands of hypotheses, and the lessons learned during this project—may help to inform the research agenda around standardized reporting in economics and could ideally strengthen the role that rigorous evidence plays in economic policy formulation.

6 Paper Exhibits

Table 1: Descriptive statistics of 317 pre-registrations as encoded in G_0

Dimension	Mean	Quantiles of outcomes				
		10th	25th	50th	75th	90th
A: Registrations						
Registered with PAP	0.4	0	0	0	1	1
Has Published Paper	0.49	0	0	0	1	1
Has Published in Top 5 Journal	0.1	0	0	0	0	0
Studies populations in LMIC	0.65	0	0	1	1	1
Specifies heterogeneity in detail	0.12	0	0	0	0	1
B: Elements of the pre-registration						
Interventions (incl. control)	3.77	2	2	3	4	7
Arms	4.79	2	2	3	5	9.4
Main Outcomes	5.36	1	2	4	6	10.4
Hypotheses (Total) per Study	15.92	2	4	9	18	32
Main Hypotheses	11.77	2	4	8	15	25.4
Primary Heterogeneity Hypotheses	4.15	0	0	0	0	4
Has more than 10 Hypotheses	0.43	0	0	0	1	1
C: Fraction of hypotheses per registration that are						
Detailed (Overall)	0.31	0	0	0	1	1
Type of hypotheses: Simple Difference	0.92	0.67	1	1	1	1
Type of hypotheses: Double Difference	0.03	0	0	0	0	0
Type of hypotheses: ‘Other’	0.05	0	0	0	0	0

Note: *Detailed* refers to whether a hypothesis is specific enough to be encoded in a precise way, without the need for a judgment call on the part of the encoder. A hypothesis is classified as “not detailed” if any of the following apply: (i) the test is not clearly specified (e.g., interventions are described but comparison arms are absent); (ii) no regression specification is provided (unless the test is otherwise obvious); (iii) the outcome is ambiguous; or (iv) the experimental design is not clearly described. LMIC above refers to low- and middle-income countries (as defined by the World Bank).

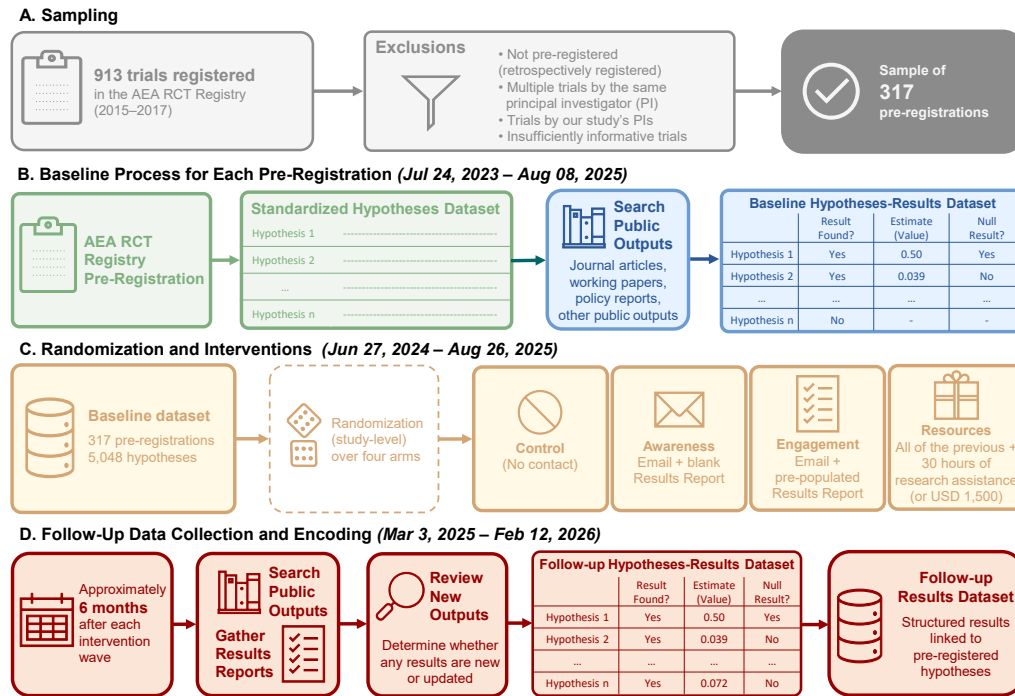


Figure 1: Study Sample, Encoding Process, and RCT Design

Note: This figure summarizes the study design and construction of the hypothesis-level datasets used throughout the paper. Panel A describes sample construction from studies registered in the AEA RCT Registry between 2015 and 2017. Panel B illustrates the baseline encoding process (July 2023–August 2025), in which each pre-registration was manually standardized into a structured hypothesis-level dataset linking interventions, outcomes, study arms, and hypotheses. Public outputs—including journal articles, working papers, and policy reports—were then systematically reviewed to identify corresponding results to each hypothesis. Panel C describes the study-level randomization and experimental interventions designed to identify the constraints limiting public reporting of pre-registered results. Intervention timing by wave is shown in Figure A20. Panel D illustrates the follow-up encoding process (March 2025–February 2026), in which newly available outputs and submitted Results Reports were reviewed and incorporated into a post-intervention hypothesis-level dataset. The same hypotheses are tracked consistently across baseline and follow-up stages of our study.

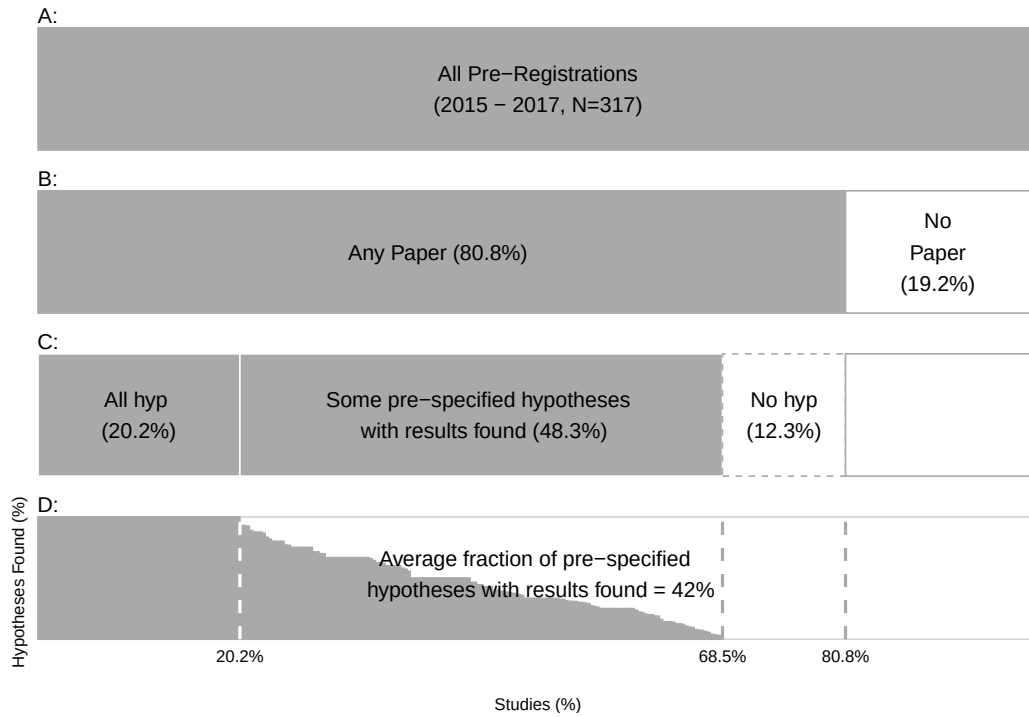


Figure 2: Reporting of Pre-specified Hypotheses in Economics

Note: The volume of the rectangle in each panel represents 100% of the sample. “Any paper” is defined in Section 4.1. “All hyp” refers to all studies’ pre-specified hypotheses whose results were found, “No hyp” means that no results corresponding to any of the pre-specified hypotheses of those studies were found. “Availability” is defined in Section 3. The height of the bottom panel represents the fraction of pre-registered hypotheses with results available by study, sorted on Available in decreasing order from left to right. The construction of the sample is described in Section 2.1.

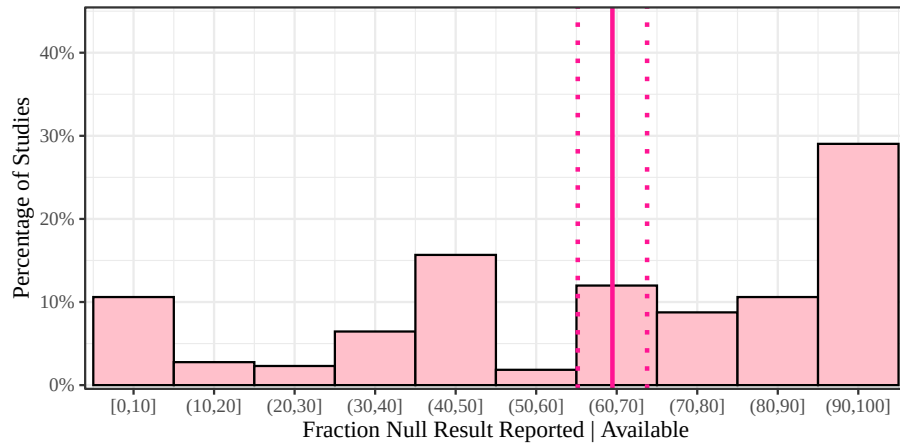


Figure 3: Fraction of Null Results Reported

Note: The histogram bars present the fraction of studies with the corresponding proportion of null results (conditional on results being available). The solid vertical line denotes the overall average fraction of null results reported (and the dotted vertical lines denote the 95% confidence interval); this proportion is 64.5%.

Table 2: Fraction of Results Available and Fraction Null Results (Conditional on Available) by Various Characteristics

Categories	Available (%)	Null Available (%)	<i>N</i>
A: Overall			
All Registrations	42.0	64.5	317
Registration with paper available	52.0	64.5	256
B: Type of publication			
Working paper	51.6	66.7	98
Published not top 5	55.1	65.5	125
Published top5	41.3	51.3	31
C: Stringency of availability			
In main body	34.8	61.1	317
No modifications	28.6	66.0	317
No mod. + main body	24.0	62.5	317
No mod + main body + not hard to find	19.7	64.3	317
Only main hypotheses	44.7	64.4	317
Study has heterogeneity:			
– All	31.4	66.0	39
– Main Only	53.8	65.4	39
– Heterogeneity Only	22.3	64.2	39
D: Other characteristics			
Studies have PAP	43.9	67.3	127
Studies in LMICs	37.8	64.3	206
Most hypotheses are clearly defined	43.0	67.4	100
More than 10 pre-registered hypotheses	31.8	68.8	137

Note: *Available* is the fraction of pre-specified hypotheses with a numerical result in any publicly accessible source. *Null | Available* is the fraction of available results with $p > 0.05$. Top-5 journals are AER, QJE, *Econometrica*, JPE, and ReStud. “No modifications” requires results to match the pre-registered hypothesis exactly; “In main body” excludes appendix-only results; “Not hard to find” reflects a subjective encoder assessment of search effort. “LMIC” refers to low- and middle-income countries (World Bank definition). “PAP” denotes a pre-analysis plan.

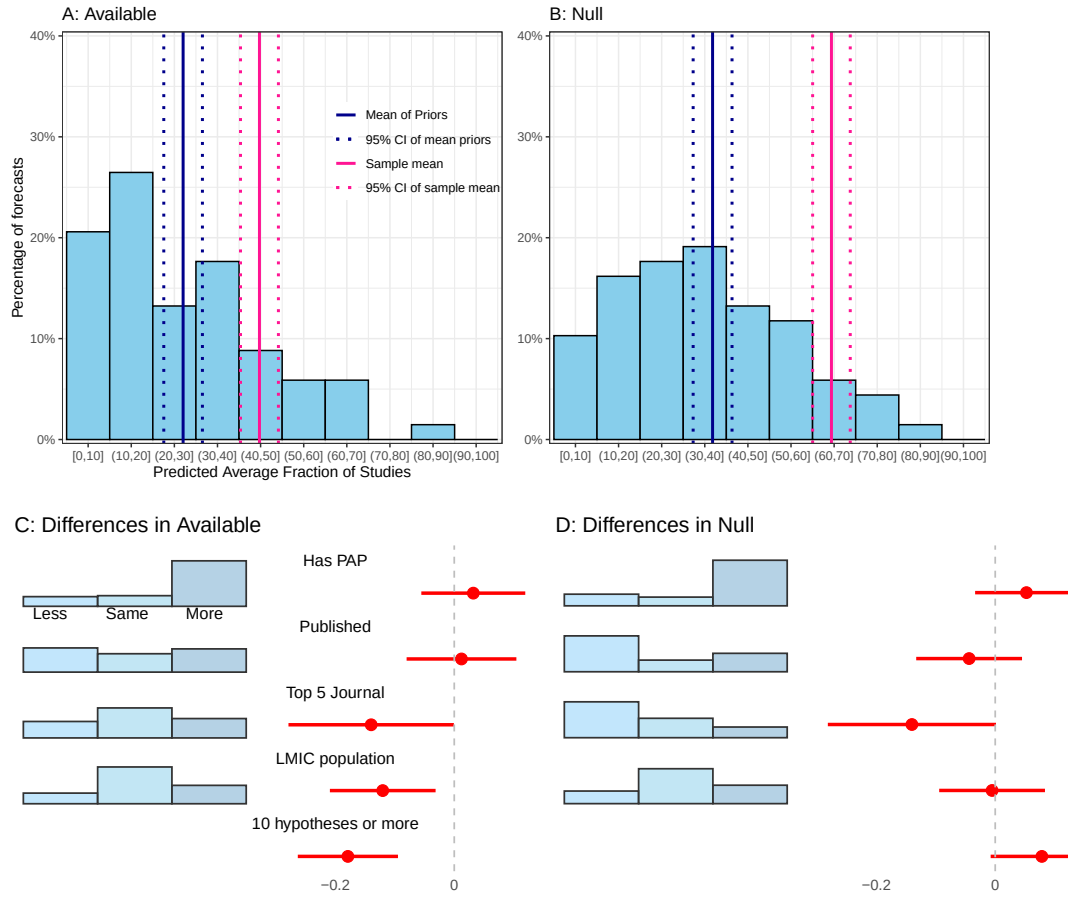


Figure 4: Priors and Data Regarding Research Results

Note: Panels A and B compare expert forecasts (bars) with realized sample means (solid pink line; 95% CI dotted) for the fraction of hypotheses available (A) and the fraction null conditional on availability (B). The solid blue line is the mean forecast. Panels C and D compare predicted and realized group differences for the same two outcomes: blue bars show the share of forecasters predicting each direction of difference; the red dot and line show the OLS point estimate and 95% CI from a bivariate regression on the subgroup indicator. *Has PAP*: registered with a pre-analysis plan. *Published*: resulted in a peer-reviewed publication. *Top 5 Journal*: AER, QJE, *Econometrica*, JPE, or ReStud. *LMIC Population*: study population in a low- or middle-income country (World Bank). *More than 10 Hypotheses*: registration contains more than 10 pre-specified hypotheses. The forecasters sample includes $N = 68$ respondents; 83.8% are economists and 76.5% are academics or graduate students.

Table 3: Engagement following the intervention email

Arm	(1) N	Clicked on:			Responded with email:		
		(2) Any (%)	(3) Results (%)	(4) Details.pdf (%)	(5) Yes (%)	(6) Informative (%)	(7) No (%)
<i>Awareness</i>	90	84.4	82.2	0.0	51.1	28.9	22.2
<i>Engagement</i>	88	92.0	92.0	52.3	76.1	60.2	15.9
<i>Resources</i>	88	90.9	90.9	47.7	69.3	50.0	19.3
Total	266	89.1	88.3	33.1	65.4	46.2	19.2

Note: Column (2) reports whether at least one author on the study clicked any link in the intervention email. Column (3) reports the click rate for the Results Report link specifically. Column (4) reports the click rate for the encoding details PDF (only sent in the Engagement and Resources arms). Columns (5)–(7) classify email responses: “Yes” includes any reply; “Informative” (6) includes replies with substantive content about the study’s results or reporting status; and “No” (7) indicates that no informative content was obtained from the email received. Fourteen (14) studies expressed interest in RA support through the Resources arm (2 via external funds and 12 via Berkeley-based RA time); in practice, six took up the Berkeley RA support and none followed up for the funds transfer.

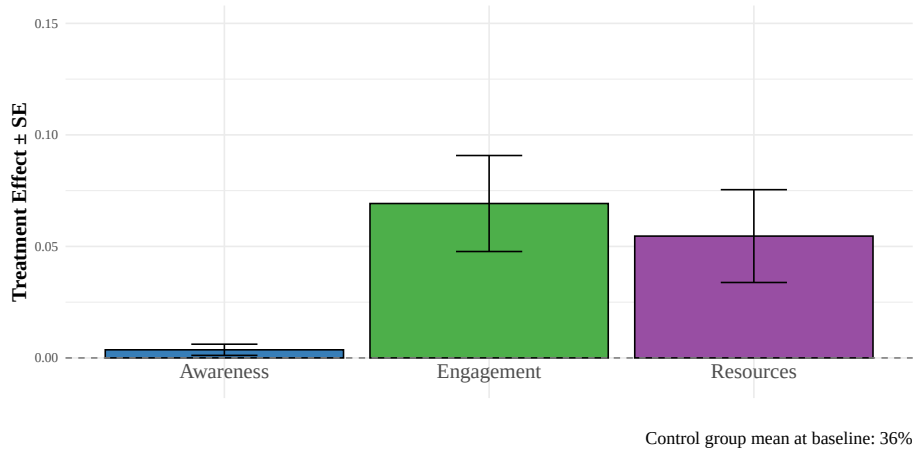


Figure 5: Treatment Effects on the fraction of available results

Note: Each bar represents the estimated treatment effect of an arm on the fraction of pre-registered hypotheses with publicly available results (*Available*), relative to the Control arm. Treatment effects are estimated using the preferred difference-in-differences specification (Equation 1). The error bars represent ± 1 standard error.

Table 4: Main Treatment Effects

	Available (DD)		Available (CS)		Explained		Fraction Missing	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Awareness	0.004 (0.003)	0.004 (0.003)	0.004 (0.007)	0.004 (0.007)	0.052 (0.023)	0.052 (0.023)	-0.056 (0.025)	-0.057 (0.026)
Engagement	0.069 (0.022)		0.066 (0.020)		0.123 (0.033)		-0.190 (0.037)	
Resources	0.055 (0.021)		0.056 (0.022)		0.083 (0.028)		-0.139 (0.035)	
Pooled (Eng. or Res.)		0.062 (0.015)		0.061 (0.015)		0.103 (0.024)		-0.164 (0.028)
Post	0.000 (0.000)	0.000 (0.000)						
Available (baseline)			0.930 (0.022)	0.929 (0.021)	-0.156 (0.034)	-0.158 (0.034)	-0.774 (0.038)	-0.771 (0.038)
<i>N</i>	634	634	317	317	317	317	317	317
Study Fixed Effects	X	X						
Controls			X	X	X	X	X	X
H1: Awareness	[0.073]		[0.268]		[0.012]		[0.013]	
H2: Engagement - Awareness	[0.001]		[0.002]		[0.025]		[0]	
H3: Resources - Engagement	[0.687]		[0.635]		[0.85]		[0.868]	

Note: *Available* is the fraction of pre-registered hypotheses for which a numerical result is found in any publicly accessible source. *Explained* is the fraction of missing hypotheses (*Available* = 0) for which the study’s authors provided a reason for non-reporting; this outcome is only observed post-intervention. *Fraction Missing* is the fraction of pre-registered hypotheses that are neither available nor explained. Columns (1)–(2) report *Available* estimated using the pre-registered difference-in-differences specification (Equation 1); columns (3)–(8) use the cross-sectional specification (Equation 2), under which the linear partition $Available + Explained + Fraction\ Missing = 1$ holds exactly across coefficients. Control group means: 0.36 (*Available*), 0 (*Explained*), 0.64 (*Fraction Missing*). The pre-specified controls are indicators for whether the study has a PAP, whether the study was undertaken in an LMIC, and indicators for each encoder on the team. The “Explained” and “Fraction Missing” outcomes were not pre-registered. Robust standard errors in parentheses; one-sided unadjusted *p*-values in square brackets. Romano-Wolf adjusted *p*-values from the pre-registered SUR family ($\{H1, H2, H3\} \times \{Available, Null\}$, 6 hypotheses; PAP §5.4) for *Available* (column (1), DD): $H1 = 0.164$, $H2 = 0.001$, $H3 = 0.746$. The joint Wald test on the six pre-registered treatment effects (three arms $\times$ the two pre-registered outcomes *Available* and *Null*; *Null* estimates are reported in Table 6, Panel A) rejects the null that they are all zero ($\chi^2(6) = 20.52$, $p = 0.002$). A companion table reporting each cross-sectional specification both with and without the baseline-*Available* control, and with the *Null* (Y_3) outcome added, is in Appendix Table A1.

Table 5: Hypotheses that become available post intervention

Type of Newly Available	Hypotheses	Studies ¹	Estimate	SE
Not Available in Baseline	24.9%	40.7%	0.025	(0.010)
Available but buried	56.1%	37.0%	0.048	(0.013)
Available but deemed inconsistent	19.0%	40.7%	0.062	(0.015)
Total	221	27		

Note: Each row adds a category of newly available hypotheses cumulatively, so the estimates are cumulative from top to bottom. *Estimate* is the pooled Engagement/Resources treatment effect on the fraction of available hypotheses; *SE* is its robust standard error. *Hypotheses* is the within-category share of all newly available hypotheses; *Studies*¹ is the share of treated studies contributing at least one hypothesis of that type. Percentages in the *Studies* column sum to more than 100% because some studies contribute hypotheses across multiple categories. An additional 96 hypotheses became available in response to the treatments (77 from 5 studies with 267 hypotheses in total) but were already available at baseline (14 due to a coding error, and 82 missed from the papers); these are not counted as newly available.

¹ Among the 27 studies with any newly available hypothesis.

Table 6: Average Percentage of Null Results Per Study, by Treatment

Panel A: Treatment Effects on Fraction Null				
	DD		CS	
	main (1)	pool (2)	main (3)	pool (4)
Awareness	0.003 (0.002)	0.003 (0.002)	-0.046 (0.034)	-0.046 (0.033)
Engagement	0.054 (0.019)		0.066 (0.037)	
Resources	0.029 (0.014)		0.041 (0.036)	
Pooled (Eng. or Res.)		0.042 (0.012)		0.053 (0.031)
Post	0.000 (0.000)	0.000 (0.000)		
Available (baseline)			0.614 (0.039)	0.613 (0.039)
<i>N</i>	634	634	317	317
Study Fixed Effects	X	X		
Controls			X	X
Baseline Avail control			X	X
H1: Awareness	[0.091]		[0.915]	
H2: Engagement - Awareness	[0.004]		[0]	
H3: Resources - Engagement	[0.85]		[0.763]	

Panel B: Descriptive Statistics						
Arm	Pre		Post			
	(1) Uncon- ditional (N = 317)	(2) Condi- tional (N = 217)	(3) Uncon- ditional (N = 317)	(4) Condi- tional (N = 225)	(5) Newly Available (N = 27)	(6) p-value (5) vs. (2)
Control	24.8	66.6	24.8	66.6		
Awareness	23.7	53.8	24.0	54.4	62.5	0.688
Engagement	28.8	69.9	34.2	70.5	75.2	0.611
Resources	31.7	69.0	34.7	67.5	60.6	0.496
Any Treatment	27.5	64.5	29.9	64.5	68.4	0.563

Note: Panel A reports treatment effects on the fraction of null results per study ($y3_study$). Columns (1)–(2) use the pre-registered DD specification (Equation 1); columns (3)–(4) use the cross-sectional specification (Equation 2) with baseline *Available* (Y_1 pre-intervention) added as a control (the ANCOVA-style robustness used in Table 4, columns (3)–(4)). Pool columns (2, 4) replace T_2 and T_3 with T_2 or T_3 . Robust standard errors in parentheses; one-sided unadjusted p -values in square brackets for H1 (Awareness arm), H2 (Engagement – Awareness) and H3 (Resources – Engagement). Romano-Wolf adjusted p -values from the pre-registered SUR family ($\{H1, H2, H3\} \times \{Available, Null\}$, 6 hypotheses; PAP §5.4) for *Null* (column (1), DD): $H1 = 0.208$, $H2 = 0.003$, $H3 = 0.826$. Panel B reports descriptive statistics. The unconditional outcome is the fraction of pre-registered hypotheses for which a null (non-significant) result was found. The conditional outcome is the fraction of (non-significant) null (non-

Table 7: Explanations Provided for Missing Results (Cumulative)

Explanation Type (each row adds the category below)	Hypo- theses	Studies	Estimate	(SE)
<i>Panel A: No data produced</i>				
Intervention failed or not implemented	18.2%	31.2%	0.055	(0.018)
+ Intervention ran, no data collected	31.2%	33.3%	0.103	(0.024)
<i>Panel B: Data collected, not reported</i>				
+ Intervention ran, data issues	6.0%	6.2%	0.118	(0.025)
+ Data available, analysis pending	20.4%	29.2%	0.130	(0.026)
+ Data available, underpowered	19.4%	10.4%	0.146	(0.028)
+ Disagree with authors	4.7%	6.2%	0.152	(0.028)
Total <i>N</i>	653	48		

Note: *Explained* is the fraction of missing hypotheses (those with *Available* = 0) for which the author provided a reason for non-reporting. *Estimate* is the pooled Engagement/Resources treatment effect on *Explained* using a modified (treatment definition) of equation 2. *Hypotheses*: within-category share of all missing hypotheses. *Studies*: share of studies providing at least one explanation of that type; sums to more than 100% because some studies gave explanations across multiple categories. An additional 8 hypotheses (3 studies) received the explanation “not pre-registered” and are excluded from all rows.

Table 8: Regression Estimates for Heterogeneity Analysis

	In Group	Study has PAP			Study is in LMIC		
		Available (1)	Explained (2)	Null (3)	Available (4)	Explained (5)	Null (6)
Awareness	No	0.003 (0.003)	0.055 (0.031)	0.003 (0.003)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
	Yes	0.005 (0.004)	0.031 (0.025)	0.003 (0.004)	0.006 (0.004)	0.069 (0.032)	0.005 (0.004)
	<i>p</i> -value for <i>Yes = No</i>	[0.765]	[0.552]	[0.961]	[0.145]	[0.032]	[0.183]
Engagement	No	0.066 (0.028)	0.102 (0.038)	0.059 (0.027)	0.078 (0.032)	0.105 (0.044)	0.053 (0.026)
	Yes	0.073 (0.034)	0.144 (0.054)	0.046 (0.027)	0.064 (0.029)	0.129 (0.044)	0.054 (0.026)
	<i>p</i> -value for <i>Yes = No</i>	[0.874]	[0.520]	[0.732]	[0.737]	[0.699]	[0.980]
Resources	No	0.006 (0.005)	0.092 (0.039)	0.009 (0.006)	0.075 (0.043)	0.075 (0.046)	0.060 (0.036)
	Yes	0.124 (0.048)	0.042 (0.029)	0.059 (0.032)	0.044 (0.022)	0.070 (0.031)	0.014 (0.009)
	<i>p</i> -value for <i>Yes = No</i>	[0.015]	[0.298]	[0.124]	[0.533]	[0.931]	[0.211]
Engagement vs. Awareness	No	[0.012]	[0.170]	[0.018]	[0.007]	[0.009]	[0.020]
	Yes	[0.047]	[0.057]	[0.112]	[0.045]	[0.272]	[0.064]
	<i>p</i> -value for <i>Yes = No</i>	[0.903]	[0.391]	[0.729]	[0.644]	[0.518]	[0.918]
Resources vs. Engagement	No	[0.983]	[0.567]	[0.967]	[0.527]	[0.678]	[0.440]
	Yes	[0.391]	[0.094]	[0.762]	[0.593]	[0.276]	[0.147]
	<i>p</i> -value for <i>Yes = No</i>	[0.092]	[0.257]	[0.208]	[0.807]	[0.730]	[0.368]

Note: Estimates are based on the heterogeneity specification (Equation 3), where subgroup effects are defined by (i) whether the study has a PAP and (ii) whether the study is conducted in an LMIC. All heterogeneity tests were pre-specified. The *Explained* outcome was not pre-specified. For *Available* and *Null*, estimates use the difference-in-differences specification (Equation 3); for *Explained*, the cross-sectional analogue is used (see Section 3). The row labeled ‘No’ reports $\hat{\tau}_j$ (the treatment effect for studies not in the subgroup); the row labeled ‘Yes’ reports $\hat{\tau}_j + \hat{\gamma}_j$ (the treatment effect for studies in the subgroup). The *p*-value for “Yes = No” tests $H_0 : \hat{\gamma}_j = 0$. Standard errors in parentheses; *p*-values in square brackets.

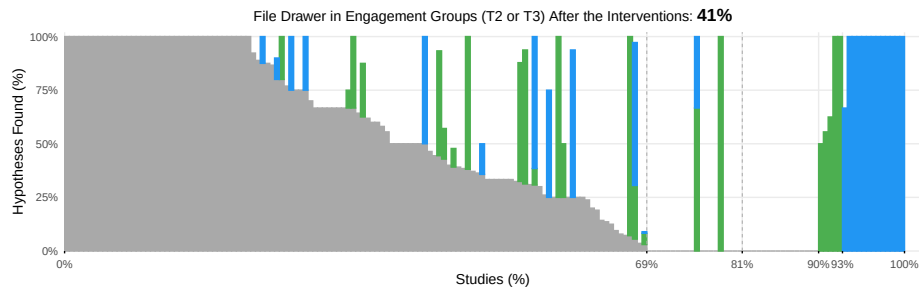


Figure 6: File Drawer After Intervention (Engagement and Resources Arms)

Note: Each bar represents one study, ordered by baseline availability. The height of each bar represents 100% of the hypotheses in a given study. Gray indicates hypotheses already available at baseline; green (darker) indicates hypotheses that became newly available post-intervention; blue indicates hypotheses that remain missing but for which the author provided an explanation. The left dashed vertical line separates studies with a paper and at least one available result (left) from studies with a paper but no available results at baseline (middle). The right dashed vertical line separates studies with a published paper (left) from studies without one (right). The bold percentage in the title reports the residual file drawer; the share of hypotheses that remain both unavailable and unexplained after the intervention.

References

- Abrams, Eliot, Jonathan Libgober, and John A List (2021). “Research registries and the credibility crisis: An empirical and theoretical investigation.”
- AEA RCT Registry (2025). “Registrations in the AEA RCT Registry (2013-05-15 through 2025-08-01).” Harvard Dataverse, doi: 10.7910/DVN/T8KQ1L.
- van den Akker, Olmo R., et al. (2023). “Selective hypothesis reporting in psychology: Comparing preregistrations and corresponding publications.” *Advances in Methods and Practices in Psychological Science*, 6(3): 25152459231187988.
- Andrews, Isaiah and Maximilian Kasy (2019). “Identification of and correction for publication bias.” *American Economic Review*, 109(8): 2766–94.
- Banerjee, Abhijit, et al. (2020). “In praise of moderation: Suggestions for the scope and use of pre-analysis plans for rcts in economics.” *NBER working paper*, (w26993).
- Brodeur, Abel, Nikolai Cook, and Anthony Heyes (2020). “Methods matter: P-hacking and publication bias in causal analysis in economics.” *American Economic Review*, 110(11): 3634–3660.
- Brodeur, Abel, et al. (2016). “Star Wars: The Empirics Strike Back.” *American Economic Journal: Applied Economics*, 8(1): 1–32.
- Brodeur, Abel, et al. (2024). “Do preregistration and preanalysis plans reduce p-hacking and publication bias? evidence from 15,992 test statistics and suggestions for improvement.” *Journal of Political Economy Microeconomics*, 2(3): 527–561.
- Cavanagh, Jack, Sarah Kopper, and Lars Vilhuber (2024). “Improving results reporting in rct registries.” Dated December 20, 2024.
- Cavanagh, Jack, et al. (2023). *A Metadata Schema for Data from Experiments in the Social Sciences*. World Bank.
- Chopra, Felix, et al. (2024). “The null result penalty.” *The Economic Journal*, 134(657): 193–219.
- Congress, U.S. (1971). “National cancer act of 1971.” Public Law 92-218, 85 Stat. 778.
- DDI Alliance (2021). *Data Documentation Initiative (DDI) Codebook 2.5: XML Schema and Documentation*. Technical report, DDI Alliance.
- De Angelis, Catherine, et al. (2004). “Clinical trial registration: a statement from the international committee of medical journal editors.” *The Lancet*, 364(9438): 911–912.
- DellaVigna, Stefano, Devin Pope, and Eva Vivalt (2019). “Predict science to improve science.” *Science*, 366(6464): 428–429.
- Edlin, Aaron S and Michael Love (2022). “Needed in empirical social science: Numbers.” *Available at SSRN 4195779*.

- Franco, Annie, Neil Malhotra, and Gabor Simonovits (2014). "Publication bias in the social sciences: Unlocking the file drawer." *Science*, 345(6203): 1502–1505.
- Hendren, Nathaniel and Ben Sprung-Keyser (2020). "A unified welfare analysis of government policies." *The Quarterly journal of economics*, 135(3): 1209–1318.
- IDEAL Consortium (2025). "The IDEAL Schema."
- Leight, Jessica, Viola Asri, and Taisuke Imai (2025). "Pathways from registration to publication: Evidence from the aea rct registry."
- Ofosu, George K and Daniel N Posner (2023). "Pre-analysis plans: An early stocktaking." *Perspectives on Politics*, 21(1): 174–190.
- Rosenthal, Robert (1979). "The file drawer problem and tolerance for null results." *Psychological bulletin*, 86(3): 638.
- Simes, Robert John (1986). "Publication bias: the case for an international registry of clinical trials." *Journal of Clinical Oncology*, 4(10): 1529–1541.
- Simmons, Joseph P., Leif D. Nelson, and Uri Simonsohn (2011). "False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant." *Psychological Science*, 22(11): 1359–1366.
- Snilstveit, Birte, et al. (2016). "Evidence & Gap Maps: A tool for promoting evidence informed policy and strategic research agendas." *Journal of Clinical Epidemiology*, 79: 120–129.
- Thibault, Robert T. and TARG Meta-Research Group (2023). "Estimating the prevalence of discrepancies between study registrations and publications: a systematic review and meta-analyses." *BMJ open*, 13(10): e076264.
- Vivalt, Eva (2020). "How much can we generalize from impact evaluations?" *Journal of the European Economic Association*, 18(6): 3045–3089.

Appendix

Figure A1: *This Project's* Results Report, Page 1 of 6

RCT Registry Results Report



Questions about how to fill this report? See this [brief explainer](#) and this [pre-filled example](#).

Hypothesis # 1 (one sentence description of the hypothesis you registered in the AEA registry)

Encouraging to post results for all pre-registered hypotheses by providing information and an empty results report template does not have an effect on the fraction of hypotheses that are completely available, including heterogeneity tests.

Please provide the closest result to this pre-registered hypothesis

$\hat{\beta}$ = 0.00365, Units: , SE = 0.00251, N = 634 , p-value = 0.0727

and/or attach document with results and provide the location below:

Page 32, Table 4 ("Main Treatment Effects"), column 1 ("Available"), row 1 ("Awarene

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

If not originally in paper/report, why:

(select all that apply)

☐ Null result

☐ Unfinished paper

☐ Other comments:

☐ Didn't collect that data

☐ Not included in write-up

Primary pre-registered heterogeneity

Dimension	Effect	SE	Comments
Study has PAP	0.00160	0.00534	p-value = 0.765
Study is in LMIC	0.00557	0.00383	p-value = 0.145

Other comments related to this hypothesis:

Heterogeneity location: Page 35, Table 8, panel Awareness, row 5, columns 1 and 4, respectively.

If more rows are needed for primary heterogeneity tests, download from [here](#).

Figure A1: *This Project's* Results Report, Page 2 of 6

RCT Registry Results Report



Questions about how to fill this report? See this [brief explainer](#) and this [pre-filled example](#).

Hypothesis # 2 (one sentence description of the hypothesis you registered in the AEA registry)

There is no incremental effect of pre-filling the results report templates relative to sending empty templates on the fraction of hypotheses that are completely available, including heterogeneity tests.

Please provide the closest result to this pre-registered hypothesis

$\hat{\beta}$ = 0.0656 , Units: , SE = 0.0217 , N = 634 , p-value = 0.0012

and/or attach document with results and provide the location below:

Page 32, Table 4 ("Main Treatment Effects"), column 1 ("Available"), row 6 ("Engagement")

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

If not originally in paper/report, why:

(select all that apply)

☐ Didn't collect that data ☐ Not included in write-up

☐ Null result ☐ Unfinished paper ☐ Other comments:

Primary pre-registered heterogeneity

Dimension	Effect	SE	Comments
Study has PAP	0.00542	0.0445	p-value = 0.903
Study is in LMIC	-0.0200	0.0433	p-value = 0.644

Other comments related to this hypothesis:

Heterogeneity location: Page 35, Table 8, panel Engagement vs. Awareness, row 3, columns 1 and 4, respectively.

If more rows are needed for primary heterogeneity tests, download from [here](#).

Figure A1: *This Project's* Results Report, Page 3 of 6

RCT Registry Results Report



CEGA
Center for Effective Global Action

Questions about how to fill this report? See this [brief explainer](#) and this [pre-filled example](#).

Hypothesis # 3 (one sentence description of the hypothesis you registered in the AEA registry)

There is no incremental effect of providing RA support and pre-filling the results report templates relative to only pre-filling the templates on the fraction of hypotheses that are completely available, including heterogeneity tests.

Please provide the closest result to this pre-registered hypothesis

$\hat{\beta}$ = -0.0146 , Units: , SE = 0.0299 , N = 634 , p-value = 0.687

and/or attach document with results and provide the location below:

Page 32, Table 4 ("Main Treatment Effects"), column 1 ("Available"), row 7 ("Resource

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

If not originally in paper/report, why:

(select all that apply)

☐

Didn't collect that data

☐

Not included in write-up

☐

Null result

☐

Unfinished paper

☐

Other comments:

Primary pre-registered heterogeneity

Dimension	Effect	SE	Comments
Study has PAP	0.111	0.0657	p-value = 0.0920
Study is in LMIC	-0.00383	0.0371	p-value = 0.807

Other comments related to this hypothesis:

Heterogeneity location: Page 35, Table 8, panel Resources vs. Engagement, row 3, columns 1 and 4, respectively.

If more rows are needed for primary heterogeneity tests, download from [here](#).

Figure A1: *This Project's* Results Report, Page 4 of 6

RCT Registry Results Report



CEGA
Center for Effective Global Action

Questions about how to fill this report? See this [brief explainer](#) and this [pre-filled example](#).

Hypothesis # 4 (one sentence description of the hypothesis you registered in the AEA registry)

Encouraging to post results for all pre-registered hypotheses by providing information and an empty results report template does not have an effect on the fraction of reported null results, including heterogeneity tests.

Please provide the closest result to this pre-registered hypothesis

$\hat{\beta}$ = 0.00312, Units: , SE = 0.00234, N = 634, p-value = 0.0912

and/or attach document with results and provide the location below:

Page 32, Table 4 ("Main Treatment Effects"), column 3 ("Null"), row 1 ("Awareness")

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

If not originally in paper/report, why:

(select all that apply)

☐

Didn't collect that data

☐

Not included in write-up

☐

Null result

☐

Unfinished paper

☐

Other comments:

Primary pre-registered heterogeneity

Dimension	Effect	SE	Comments
Study has PAP	0.000235	0.00481	p-value = 0.961
Study is in LMIC	0.00476	0.00357	p-value = 0.183

Other comments related to this hypothesis:

Heterogeneity location: Page 35, Table 8, panel Awareness, row 5, columns 3 and 6, respectively.

If more rows are needed for primary heterogeneity tests, download from [here](#).

Figure A1: *This Project's* Results Report, Page 5 of 6

RCT Registry Results Report



CEGA
Center for Effective Global Action

Questions about how to fill this report? See this [brief explainer](#) and this [pre-filled example](#).

Hypothesis # 5 (one sentence description of the hypothesis you registered in the AEA registry)

There is no incremental effect of pre-filling the results report templates relative to sending empty templates on the fraction of reported null results, including heterogeneity tests.

Please provide the closest result to this pre-registered hypothesis

$\hat{\beta}$ = 0.0506 , Units: , SE = 0.0192 , N = 634 , p-value = 0.00414

and/or attach document with results and provide the location below:

Page 32, Table 4 ("Main Treatment Effects"), column 3 ("Null"), row 6 ("Engagement -

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

If not originally in paper/report, why:

(select all that apply)

☐

Didn't collect that data

☐

Not included in write-up

☐

Null result

☐

Unfinished paper

☐

Other comments:

Primary pre-registered heterogeneity

Dimension	Effect	SE	Comments
Study has PAP	-0.0132	0.0381	p-value = 0.729
Study is in LMIC	-0.00383	0.0371	p-value = 0.918

Other comments related to this hypothesis:

Heterogeneity location: Page 35, Table 8, panel Engagement vs. Awareness, row 3, columns 3 and 6, respectively.

If more rows are needed for primary heterogeneity tests, download from [here](#).

Figure A1: *This Project's* Results Report, Page 6 of 6

RCT Registry Results Report



CEGA
Center for Effective Global Action

Questions about how to fill this report? See this [brief explainer](#) and this [pre-filled example](#).

Hypothesis # 6 (one sentence description of the hypothesis you registered in the AEA registry)

There is no incremental effect of providing RA support and pre-filling the results report templates relative to only pre-filling the templates on the fraction of reported null results, including heterogeneity tests.

Please provide the closest result to this pre-registered hypothesis

$\hat{\beta} = -0.0243$, Units: , SE = 0.0234, N = 634, p-value = 0.850

and/or attach document with results and provide the location below:

Page 32, Table 4 ("Main Treatment Effects"), column 3 ("Null"), row 7 ("Resources - E

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

If not originally in paper/report, why:

(select all that apply)

☐

Didn't collect that data

☐

Not included in write-up

☐

Null result

☐

Unfinished paper

☐

Other comments:

Primary pre-registered heterogeneity

Dimension	Effect	SE	Comments
Study has PAP	0.0626	0.0497	p-value = 0.208
Study is in LMIC	-0.0470	0.0522	p-value = 0.368

Other comments related to this hypothesis:

Heterogeneity location: Page 35, Table 8, panel Resources vs. Engagement, row 3, columns 3 and 6, respectively.

If more rows are needed for primary heterogeneity tests, download from [here](#).

Figure A2: *Awareness* Treatment Sample Email

Treatment #1

Subject: Reporting results for your study on the AEA RCT Registry

Dear {{Researcher Name}},

We are a team of researchers studying pre-registration, pre-analysis plans, and the reporting of results in social science research, funded by the Research Council of Norway (#262675) and the Berkeley Initiative for Transparency in the Social Sciences (BITSS).

To have an accurate representation of evidence in our field, it's essential to keep track of the results from pre-registered studies, regardless of how they turn out. For that purpose, we've developed a short **results report** to standardize and streamline the reporting of all research hypotheses for studies on the AEA Registry.

The filled-in results report for your study will serve as a useful resource for the economics research community. We encourage you to upload a results report from your study “{{TITLE OF STUDY}}” to the AEA Registry. In the form, we ask you to briefly describe **the hypotheses recorded in {{AEA-ID}} and their associated results**. To help you navigate this process, we are linking [an empty results report](#), as well as a [pre-filled example](#) from another study. The linked results report template is a suggested format, but of course, you are free to share the results in other formats, including in a paper. To submit, please upload via the Post-Trial section on the AEA RCT Registry page for your study ([in the “Reports, Papers, and Other Materials” tab](#)), or if you prefer, you can attach the file in a reply to this email.

We plan to use the information you and other scholars report in a study of publication patterns in economics. As a part of this project, study-level information from the results reports – including project information you share with us, such as pre-analysis plans or additional statistical results – may be shared publicly online (though we will not highlight individual projects or authors). Please review the [study consent form](#), and feel free to reach out with any questions.

We understand that providing this information is a non-trivial ask, and we would be very grateful for your participation.

Many thanks,

Edward Miguel (UC Berkeley)
Bertil Tungodden (NHH Norwegian School of Economics)
Erik Ø. Sørensen (NHH Norwegian School of Economics)
Fernando Hoces de la Guardia (UC Berkeley)

Figure A3: *Engagement* Treatment Sample Email

Treatment #2

Subject: Results report for your study on the AEA RCT Registry

Dear {{Researcher Name}},

We are a team of researchers studying pre-registration, pre-analysis plans, and the reporting of results in social science research, funded by the Research Council of Norway (#262675) and the Berkeley Initiative for Transparency in the Social Sciences (BITSS).

To have an accurate representation of evidence in our field, it's essential to keep track of the results from pre-registered studies, regardless of how they turn out. For that purpose, we've developed a short **results report** to standardize and streamline the reporting of all research hypotheses for studies on the AEA Registry.

Our team has read the materials on the AEA Registry for your study “{{TITLE OF STUDY}}” ({{AEA-ID}}) and encoded the research hypotheses into the {{Hyperlink}}. We've also read any paper(s) associated with this registration and attempted to record the results of the primary hypotheses. **If possible, we ask you to review these materials and provide feedback in two areas:**

1. Did we report the hypotheses (as you recorded them on the AEA Registry) and the results of your study correctly? If you disagree or would have encoded them differently, please note your preferred hypotheses and results using the linked results report.
2. In some cases, we could not find the results for specific research hypotheses recorded on the AEA Registry. Could you fill in these estimates or report why they are not available?

The filled-in results report for your study will serve as a useful resource for the economics research community. We encourage you to upload a results report from your study “{{TITLE OF STUDY}}” to the AEA Registry. To help you navigate this process, we are linking [a pre-filled example results report](#) from another study. The linked results report template is a suggested format, but of course, you are free to share the results in other formats, including in a paper. To submit, please upload via the Post-Trial section on the AEA RCT Registry page for your study ([in the “Reports, Papers, and Other Materials” tab](#)), or if you prefer, you can attach the file in a reply to this email.

We plan to use the information you and other scholars report in a study of publication patterns in economics. As a part of this project, study-level information from the results reports – including project information you share with us, such as pre-analysis plans or additional statistical results – may be shared publicly online (though we will not highlight individual projects or authors). Please review the [study consent form](#), and feel free to reach out with any questions.

We understand that providing this information is a non-trivial ask, and we would be very grateful for your participation.

Many thanks,

Edward Miguel (UC Berkeley)
Bertil Tungodden (NHH Norwegian School of Economics)
Erik Ø. Sørensen (NHH Norwegian School of Economics)
Fernando Hoces de la Guardia (UC Berkeley)

Figure A4: *Resources* Treatment Sample Email

Treatment #3

Subject: Results report for your study on the AEA RCT Registry + RA support

Dear {{Researcher Name}},

We are a team of researchers studying pre-registration, pre-analysis plans, and the reporting of results in social science research, funded by the Research Council of Norway (#262675) and the Berkeley Initiative for Transparency in the Social Sciences (BITSS).

To have an accurate representation of evidence in our field, it's essential to keep track of the results from pre-registered studies, regardless of how they turn out. For that purpose, we've developed a short **results report** to standardize and streamline the reporting of all research hypotheses for studies on the AEA Registry.

Our team has read the materials on the AEA Registry for your study "{{TITLE OF STUDY}}" ({{AEA-ID}}) and encoded the research hypotheses into the {{Hyperlink}}. We've also read any paper(s) associated with this registration and attempted to record the results of the primary hypotheses. **If possible, we ask you to review these materials and provide feedback in two areas:**

1. Did we report the hypotheses (as you recorded them on the AEA Registry) and the results of your study correctly? If you disagree or would have encoded them differently, please note your preferred hypotheses and results using the linked results report.
2. In some cases, we could not find the results for specific research hypotheses recorded on the AEA Registry. Could you fill in these estimates or report why they are not available?

The filled-in results report for your study will serve as a useful resource for the economics research community. We encourage you to upload a results report from your study "{{TITLE OF STUDY}}" to the AEA Registry. To help you navigate this process, we are linking [a pre-filled example results report](#) from another study. The linked results report template is a suggested format, but of course, you are free to share the results in other formats, including in a paper. To submit, please upload via the Post-Trial section on the AEA RCT Registry page for your study ([in the "Reports, Papers, and Other Materials" tab](#)), or if you prefer, you can attach the file in a reply to this email.

To facilitate this work, we are able to offer you the support of a UC Berkeley-based research assistant (at no charge to you) who has knowledge of common programming software and languages (e.g., STATA and R). We can offer you up to 30 hours of RA work time on this task of obtaining study estimates, specifically, on data cleaning, analysis, and documentation. Alternatively, if this proves to be a logistical challenge, we can offer **up to \$1,500 to cover the RA time required on your end to find or produce the estimates for the results report**. Please let us know if you are interested in this support by responding to this email (ucbitss@berkeley.edu) in the next 14 days (by {{Date}}).

We plan to use the information you and other scholars report in a study of publication patterns in economics. As a part of this project, study-level information from the results reports – including project information you share with us, such as pre-analysis plans or additional statistical results – may be shared publicly online (though we will not highlight individual projects or authors). Please review the [study consent form](#), and feel free to reach out with any questions.

We understand that providing this information is a non-trivial ask, and we would be very grateful for your participation.

Many thanks,

Edward Miguel (UC Berkeley)
Bertil Tungodden (NHH Norwegian School of Economics)
Erik Ø. Sørensen (NHH Norwegian School of Economics)
Fernando Hoces de la Guardia (UC Berkeley)

Figure A5: Awareness Treatment's Study Consent Form



CONSENT TO PARTICIPATE IN RESEARCH

Pre-Registration, Reporting Guidelines and Publication Patterns in Economics

Introduction: My name is Fernando Hoces De La Guardia and I am working with Edward Miguel, a Professor of Economics at the University of California, Berkeley to conduct a research study, which we invite you to take part in.

Purpose: The purpose of this research study is to *learn about publication patterns in the social sciences through the use of results reports*.

Procedures: If you agree to be in this study, you will be sent a results report template over email with a brief message on the importance of reporting results from pre-registered studies, regardless of how they turn out, and encouraging you to fill up and upload the results report to the AEA Registry. The time it will take to populate and upload this results report is variable. All activities can take place remotely.

Benefits: There are no direct benefits to participating in this study. We hope that the information gained from the study will help improve understanding of results reporting and publication patterns in the social sciences.

Risks/Discomforts: Given that study-level data may be shared publicly, there is a chance that the level of adherence of a published study to a pre-analysis plan could be linked to you and your co-authors.

Future use of study data: Study-level data, including non-public sources or elements you share with us, may be shared publicly online.

Compensation and Costs: You will not receive any monetary compensation for your participation in this study, nor will you be charged for any of the study activities.

Rights: Participation in research is completely voluntary. You have the right to decline to participate or to withdraw at any point in this study without penalty or loss of benefits to which you are otherwise entitled.

Questions: If you have any questions or concerns about this study, you may contact Professor Edward Miguel at emiguel@berkeley.edu. If you have any questions or concerns about your rights and treatment as a research subject, you may contact the office of UC Berkeley's Committee for the Protection of Human Subjects, at 510-642-7461 or subjects@berkeley.edu.

If you do not agree to participate, please reply to this email letting us know.

CPHS # 2023-08-16633Page 1 of 1

Figure A6: *Engagement* Treatment's Study Consent Form, Page 1 of 2



CONSENT TO PARTICIPATE IN RESEARCH

Pre-Registration, Reporting Guidelines and Publication Patterns in Economics

Introduction: My name is Fernando Hoces De La Guardia and I am working with Edward Miguel, a Professor of Economics at the University of California, Berkeley to conduct a research study, which we invite you to take part in.

Purpose: The purpose of this research study is *to learn about publication patterns in the social sciences through the use of results reports*.

Procedures: If you agree to be in this study, you will be sent a results report template over email pre-populated with information from your study's AEA Registry page with a brief message on the importance of reporting results from pre-registered studies, regardless of how they turn out, and encouraging you to upload the report to the AEA Registry. The time it will take to review and upload this results report is variable. All activities can take place remotely.

Benefits: There are no direct benefits to participating in this study. We hope that the information gained from the study will help improve understanding of results reporting and publication patterns in the social sciences.

Risks/Discomforts: Given that study-level data may be shared publicly, there is a chance that the level of adherence of a published study to a pre-analysis plan could be linked to you and your co-authors.

Future use of study data: Study-level data, including non-public sources or elements you share with us, may be shared publicly online.

Compensation and Costs: You will not receive any monetary compensation for your participation in this study, nor will you be charged for any of the study activities.

Rights: Participation in research is completely voluntary. You have the right to decline to participate or to withdraw at any point in this study without penalty or loss of benefits to which you are otherwise entitled.

Questions: If you have any questions or concerns about this study, you may contact Professor Edward Miguel at emiguel@berkeley.edu. If you have any questions or concerns

CPHS # 2023-08-16633 Page 1 of 2

Figure A6: *Engagement* Treatment's Study Consent Form, Page 2 of 2

about your rights and treatment as a research subject, you may contact the office of UC Berkeley's Committee for the Protection of Human Subjects, at 510-642-7461 or subjects@berkeley.edu.

If you do not agree to participate, please reply to this email letting us know.

Figure A7: *Resources* Treatment's Study Consent Form, Page 1 of 2

CONSENT TO PARTICIPATE IN RESEARCH

Pre-Registration, Reporting Guidelines and Publication Patterns in Economics

Introduction: My name is Fernando Hoces De La Guardia and I am working with Edward Miguel, a Professor of Economics at the University of California, Berkeley to conduct a research study, which we invite you to take part in.

Purpose: The purpose of this research study is *to learn about publication patterns in the social sciences through the use of results reports.*

Procedures: If you agree to be in this study, you will be sent a results report template over email pre-populated with information from your study's AEA Registry page with a brief message on the importance of reporting results from pre-registered studies, regardless of how they turn out, and encouraging you to upload the results report to the AEA Registry. You will also be offered up to 30 hours of RA support to obtain the study estimates. Alternatively, if this proves to be a logistical challenge, we can offer up to \$1,500 to cover the RA time on your end. The time it takes to coordinate with the RAs, review the output, and upload the report is variable. All activities can take place remotely.

Benefits: You will be offered up to 30 hours of RA support to obtain the study estimates. Alternatively, if this proves to be a logistical challenge, we can offer up to \$1,500 to cover the RA time on your end. We hope that the information gained from the study will help improve understanding of results reporting and publication patterns in the social sciences.

Risks/Discomforts: Given that study-level data may be shared publicly, there is a chance that the level of adherence of a published study to a pre-analysis plan could be linked to you and your co-authors.

Future use of study data: Study-level data, including non-public sources or elements you share with us, may be shared publicly online.

Compensation and Costs: You will not receive any monetary compensation for your participation in this study, nor will you be charged for any of the study activities.

Figure A7: *Resources* Treatment's Study Consent Form, Page 2 of 2



Rights: Participation in research is completely voluntary. You have the right to decline to participate or to withdraw at any point in this study without penalty or loss of benefits to which you are otherwise entitled.

Questions: If you have any questions or concerns about this study, you may contact Professor Edward Miguel at emiguel@berkeley.edu. If you have any questions or concerns about your rights and treatment as a research subject, you may contact the office of UC Berkeley's Committee for the Protection of Human Subjects, at 510-642-7461 or subjects@berkeley.edu.

If you do not agree to participate, please reply to this email letting us know.

Figure A8: *Empty* Results Report Sample Cover



Reporting results for your study on the AEA RCT Registry

We are a team of researchers studying pre-registration, pre-analysis plans, and the reporting of results in social science research. Our research team has developed a short template to facilitate reporting all pre-registered hypotheses for a given study.

It would be extremely valuable for the research community if you could provide results.

We also encourage you to post these results on the AEA RCT Registry page for your study (in the ["Reports, Papers, and Other Materials" section](#)).

We thank you for your valuable participation.

Edward Miguel (UC Berkeley)
Bertil Tungodden (NHH Norwegian School of Economics)
Erik Ø. Sørensen (NHH Norwegian School of Economics)
Fernando Hoces de la Guardia (UC Berkeley)

Figure A9: Empty Results Report Sample Page

RCT Registry Results Report





Questions about how to fill this report? See this [brief explainer](#) and this [pre-filled example](#).

Hypothesis # (one sentence description of the hypothesis you registered in the AEA registry)

Please provide the closest result to this pre-registered hypothesis

$\hat{\beta}$ = , Units: , SE = , N = , p-value =

and/or attach document with results and provide the location below:

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

If not originally in paper/report, why: ☐ Didn't collect that data ☐ Not included in write-up
(select all that apply)

☐ Null result ☐ Unfinished paper ☐ Other comments:

Primary pre-registered heterogeneity

Dimension	Effect	SE	Comments

Other comments related to this hypothesis:

If more rows are needed for primary heterogeneity tests, download from [here](#).

Note: Sent to all three treatment arms.

Figure A10: *Pre-Filled* Results Report Sample Cover



Reporting results for your study on the AEA RCT Registry:

Study title goes here

We are a team of researchers studying pre-registration, pre-analysis plans, and the reporting of results in social science research. Our research team has developed a short template to facilitate reporting all pre-registered hypotheses for a given study.

We have extracted hypotheses from your study registered on the AEA RCT registry and attempted to find results for each of them.

We extracted x main hypotheses from the registration. We found results for y main hypotheses.

It would be extremely valuable for the research community if you could provide results that we were unable to find and review the ones that we found. We also encourage you to post these results via the Post-Trial section on the AEA RCT Registry page for your study (in the [“Reports, Papers, and Other Materials” tab](#)). If you would like to use an empty results report template, you can access it by [clicking here](#).

We thank you for your valuable participation.

Edward Miguel (UC Berkeley)
Bertil Tungodden (NHH Norwegian School of Economics)
Erik Ø. Sørensen (NHH Norwegian School of Economics)
Fernando Hoces de la Guardia (UC Berkeley)

Note: Sent only to *Engagement* and *Resources* treatment arms.

Figure A11: Found Results Report Sample Page

RCT Registry Results Report

Main Results Found

BITSS

FAIR

CEGA

Centre for Experimental Research on Fairness, Inequality and Rationality

Center for Effective Global Action

Questions about how to fill in this report? See this [brief explainer](#) and this [pre-filled example](#).

[Here](#) is your original registration, and the attachment [details_lastname.pdf](#) contains the details of how your registration was encoded.

Hypothesis # 10

(as interpreted from pre-registration)

Outcome 5 is not different in Treatment 3 relative to Treatment 2.

Mostly agree with this statement?

☐ Yes ☐ No

If not, please say why and add correct one:

Closest result to this pre-registered hypothesis

(as found in article, publication, or write-up)

$\hat{\beta} = 12345$, SE = 54321

Location: Page X, Table Y, col W, row Z

Agree with result?

☐ Yes ☐ No

If not, please tell us why here:

Primary pre-registered heterogeneity

Agree with below? ☐ Yes ☐ No

If you disagree with this interpretation of estimates and/or believe that some dimensions of primary pre-registered heterogeneity are missing, add or modify them below. We will keep track of your edits.

Dimension	Effect	SE	Comments
Heterogeneity A	12345	12345	
Heterogeneity B	12345	12345	

If more rows are needed for primary heterogeneity tests, download from [here](#).

Note: Sent only to Engagement and Resources treatment arms.

Figure A12: *Not Found* Results Report Sample Page

RCT Registry Results Report

Main Results **Not Found**



Center for Effective Global Action

Questions about how to fill in this report? See this [brief explainer](#) and this [pre-filled example](#).

[Here](#) is your original registration, and the attachment [details_lastname.pdf](#) contains the details of how your registration was encoded.

Hypothesis # 10 (as interpreted from pre-registration)

Outcome 5 is not different in Treatment 3 relative to Treatment 2.

Mostly agree with this statement? ☐ Yes ☐ No If not, please say why and add correct one:

Please provide closest result to this pre-registered hypothesis. *We could not find this result.*

$\hat{\beta}$ = , Units: , SE = , N = , p-value =

and/or attach document with results and provide the location below:

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

Why not originally in paper/report:
(select all that apply)

☐ Didn't collect that data ☐ Not included in write-up
☐ Null result ☐ Unfinished paper ☐ Missed by reviewer team

Other comments:

Primary pre-registered heterogeneity Agree with below? ☐ Yes ☐ No

If you disagree with this interpretation of estimates and/or believe that some dimensions of primary pre-registered heterogeneity are missing, add or modify them below. We will keep track of your edits.

Dimension	Effect	SE	Comments
Heterogeneity A	Not found	Not found	
Heterogeneity B	Not found	Not found	

If more rows are needed for primary heterogeneity tests, download from [here](#).

Note: Sent only to *Engagement* and *Resources* treatment arms.

Figure A13: Extra Heterogeneity Rows Sample Page

Questions about how to fill in this report card? See this [brief explainer](#) and this [pre-filled example](#).



Additional Primary Heterogeneity Tests

Hypothesis #

Dimension	Effect	SE	Comments

Figure A14: *Not Found* Results Report Sample Page (as Filled by Authors)

RCT Registry Results Report

Main Results **Not Found**



Questions about how to fill in this report? See this [brief explainer](#) and this [pre-filled example](#).

[Here](#) is your original registration, and the attachment [details_lastname.pdf](#) contains the details of how your registration was encoded.

Hypothesis # 10 (as interpreted from pre-registration)

Outcome 5 is not different in Treatment 3 relative to Treatment 2.

Mostly agree with this statement? ☒ Yes ☐ No If not, please say why and add correct one:

Please provide closest result to this pre-registered hypothesis. *We could not find this result.*

$\hat{\beta}$ = 12345 , Units: kg/ha , SE = 54321 , N = 12345 , p-value = 54321

and/or attach document with results and provide the location below:

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

Why not originally in paper/report:
(select all that apply)

☐

Didn't collect that data

☒

Not included in write-up

☐

Null result

☐

Unfinished paper

☐

Missed by reviewer team

Other comments: Authors would write here the reason why it was not included in the write-up

Primary pre-registered heterogeneity

Agree with below? ☒ Yes ☐ No

If you disagree with this interpretation of estimates and/or believe that some dimensions of primary pre-registered heterogeneity are missing, add or modify them below. We will keep track of your edits.

Dimension	Effect	SE	Comments
Heterogeneity A	12345	12345	
Heterogeneity B	12345	12345	

If more rows are needed for primary heterogeneity tests, download from [here](#).

Note: Sent only to *Engagement* and *Resources* treatment arms.







RCT Registry Results Report
Main Results **Found**

Internal numbering

Questions about how to fill in this report? See the [brief explainer](#).
[Here](#) is your original registration, and the attachment [Detalle_Miguel.pdf](#) contains the details of how your registration was encoded.





Hypothesis

This box provides a brief description of a main hypotheses pre-registered in the AEA registry. For a detailed breakdown of all information related to each hypothesis, please refer to the accompanying file sent with this report.

Closest result to pre-registered hypothesis

The section below presents our assessment of the closest result matching the pre-registered hypothesis. If you disagree with this assessment, please indicate this and provide the closest estimate in the box below.

Closest result to this pre-registered hypothesis (as found in article, publication, or write-up)

$\hat{\beta} = 239$, SE = 129 , p-value = 0.069

Location: Online Appendix, Page 9, Table S3, column 1, panel C

Agree with result? ☒ Yes ☐ No **If not, please tell us why here:**

A better location is Table 1, Column 1, Panel C

Primary pre-registered heterogeneity

The last section outlines any pre-registered heterogeneity sub-hypotheses that we assessed were central to the study and pre-specified in sufficient detail.

If this section is empty, it indicates that we assessed that no pre-registered heterogeneity sub-hypothesis was primary. You are welcome to provide additional results if you view them as primary hypotheses in your pre-registration. If more rows are needed, please use [this link](#) to download a template with additional rows. Please ensure that the new page is numbered with the same hypothesis number as indicated on the top left of this page for easy reference. You can use the "Comment" box to provide any additional details or results.

Primary pre-registered heterogeneity Agree with below? ☒ Yes ☐ No

If you disagree with this interpretation of estimates and/or believe that some dimensions of primary pre-registered heterogeneity are missing, add or modify them below. We will keep track of your edits.

Dimension	Effect	SE	Comments
Gender (Female)	36	107	Tests gender subgroups separately
Gender (Male)	439	252	Tests gender subgroups separately

If more rows are needed for primary heterogeneity tests, download from [here](#)

Figure A16: *Not Found* Results Report Explainer





Results Report Explainer | Results **Not Found**

RCT Registry Results Report
Main Results and Heterogeneity *Not Found*

Internal numbering Questions about how to fill in this report? See this [brief explainer](#).
 Here is your original registration, and the attached [details_Miguel.pdf](#) contains the details of how your registration was encoded.

Hypothesis # 1 (as interpreted from pre-registration)

Pooled analysis across KLPS-2, KLPS-3, and KLPS-4 shows no significant difference in annual total earnings between early and later PSDP program recipients

Mostly agree with this statement? ☒ Yes ☐ No If not, please say why and add correct one:

Hypothesis

This box provides a brief description of a main hypotheses pre-registered in the AEA registry. For a detailed breakdown of all information related to each hypothesis, please refer to the accompanying **file** sent with this report.

Closest result to this pre-registered hypothesis: *We could not find this result.*

The section below should present the closest result to the hypothesis as pre-registered. We were unable to locate a report of this result that matches the specification of the pre-registered hypothesis. If possible, please assist us by providing the result of this pre-specified hypothesis, either by attaching a paper or report and indicating the location of the result within the document, and/or entering the result in the provided boxes.

You can input the estimate, outcome units, standard error, and p-value for the hypothesis here. If the estimates are unavailable, you can note qualitative results (null or significant). The report card also asks about authors' reporting decisions. You can select multiple reasons and/or provide a free-text explanation.

Please provide closest result to this pre-registered hypothesis. *We could not find this result.*

β = 80 , Units: 2017 US\$, SE = 76 , N = 13,624 , p-value = 0.297

and/or attach document with results and provide the location below:
[Hamory et al. \(2021\), Table 1, Panel B, column 1](#)

Can't access it right now, but I remember it was ☐ statistically significant ☐ null

Why not originally in paper/report: ☐ Didn't collect that data ☐ Not included in write-up
(select all that apply)

☐ Null result ☐ Unfinished paper ☒ Missed by reviewer team

Other comments:

Primary pre-registered heterogeneity

The last section outlines any pre-registered heterogeneity sub-hypotheses that we assessed were central to the study and pre-specified in sufficient detail.

If this section is empty, it indicates that we assessed that no pre-registered heterogeneity sub-hypothesis was primary. You are welcome to provide additional results if you view them as primary hypotheses in your pre-registration. You can use the "Comment" box to provide any additional details or results. If more rows are needed, please use [this link](#) to download a template with additional rows.

Primary pre-registered heterogeneity Agree with below? ☐ Yes ☒ No

If you disagree with this interpretation of estimates and/or believe that some dimensions of primary pre-registered heterogeneity are missing, add or modify them below. We will keep track of your edits.

Dimension	Effect	SE	Comments
Gender (Female)	41	62	Table 1, Panel B
Gender (Male)	118	133	Table 1, Panel B

If more rows are needed for primary heterogeneity tests, download from [here](#).

If a dimension is listed but no results are included, it means that we identified a pre-registered primary heterogeneity dimension but were unable to locate its corresponding results. You can enter the missing information in the corresponding row (and remove the label "Not found").

Figure A17: Summary Encoded Pre-Registration (G_0) Sample. Page 1 of 2. Each study in Engagement and Resources arm received a version of their study in this format under the link “details-lastname.pdf” at the top of each Results Report page.

AEA RCT Trial Registration Summary

0000

Title: The title goes here

<https://www.socialscisearch.org/trials/0000>

1 Populations

	label	country	Randomization	N	coverage
1	population1	GLO	Insert unit here	12345	The description of population 1 goes here

2 Outcomes

	label	unit_original	unit_analytical	description
1	outcome1	Insert unit here	Insert unit here	The description of outcome 1 goes here
2	outcome2	Insert unit here	Insert unit here	The description of outcome 2 goes here
3	outcome3	Insert unit here	Insert unit here	The description of outcome 3 goes here
4	outcome4	Insert unit here	Insert unit here	The description of outcome 4 goes here
5	outcome5	Insert unit here	Insert unit here	The description of outcome 5 goes here

3 Interventions

	label	description
1	intervention1	The description of intervention 1 goes here
2	intervention2	The description of intervention 2 goes here
3	intervention3	The description of intervention 3 goes here

4 Arms

	label	population	intervention
1	control	population1	NaN
2	treatment1	population1	intervention1
3	treatment2	population1	intervention2
4	treatment3	population1	intervention3

5 Arm Groups

label	armlabel
-------	----------

Figure A17: Encoded Pre-Registration (G_0) Sample, Page 2 of 2

6 Hypotheses

	label	description
1	hypothesis1	The description of hypothesis 1 goes here $E[outcome1 treatment1] - E[outcome1 control] = 0$
2	hypothesis2	The description of hypothesis 2 goes here $E[outcome2 treatment1] - E[outcome2 control] = 0$
3	hypothesis3	The description of hypothesis 3 goes here $E[outcome3 treatment1] - E[outcome3 control] = 0$
4	hypothesis4	The description of hypothesis 4 goes here $E[outcome4 treatment1] - E[outcome4 control] = 0$
5	hypothesis5	The description of hypothesis 5 goes here $E[outcome5 treatment1] - E[outcome5 control] = 0$
6	hypothesis6	The description of hypothesis 6 goes here $E[outcome1 treatment3] - E[outcome1 treatment2] = 0$
7	hypothesis7	The description of hypothesis 7 goes here $E[outcome2 treatment3] - E[outcome2 treatment2] = 0$
8	hypothesis8	The description of hypothesis 8 goes here $E[outcome3 treatment3] - E[outcome3 treatment2] = 0$
9	hypothesis9	The description of hypothesis 9 goes here $E[outcome4 treatment3] - E[outcome4 treatment2] = 0$
10	hypothesis10	The description of hypothesis 10 goes here $E[outcome5 treatment3] - E[outcome5 treatment2] = 0$

7 Heterogeneity

	subgroups	effect_type	hypothesis_id
1	Heterogeneity A	different_effects	hypothesis1, hypothesis2, hypothesis3, hypothesis4, hypothesis5, hypothesis6, hypothesis7, hypothesis8, hypothesis9, hypothesis10
2	Heterogeneity B	different_effects	hypothesis1, hypothesis2, hypothesis3, hypothesis4, hypothesis5, hypothesis6, hypothesis7, hypothesis8, hypothesis9, hypothesis10

8 Judgment Calls

	concerning	call
1	hypotheses	The description of the judgment call goes here

Figure A18: AEA RCT Registry Post Section Update Explainer



Post Trial Update Explainer

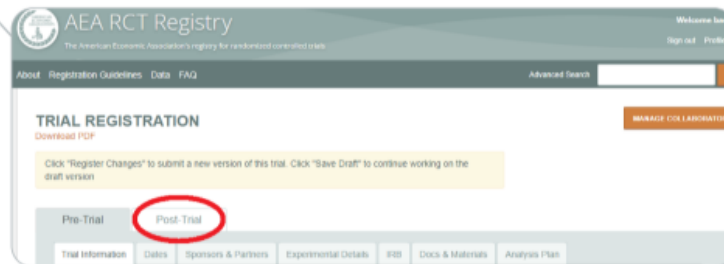


1. Open your preferred browser and go to <https://www.socialscisearch.org/trials/>.
2. Once you log in with your credentials, click on "My Trials" on the upper-right-hand-corner.
3. Click on the "Registered Trials" tab (see screenshot).
4. Select the study for which you wish to upload the results report. The corresponding page of the selected study should open.

5. On the right side, click on "Edit Trial". An editable version of the "Trial Registration" page should open.
6. Select the "Post-Trial" tab (see screenshot).

If the "Post-Trial" tab is **not** selectable, follow these steps:

- (i). Under the "Pre-Trial" tab & "Dates" subtab, ensure that the "Trial End Date" is in the past (i.e. prior to current date)
- (ii). Now click on the "Pre-Trial" tab & "Trial Information" subtab. Under the "Additional Trial Information" section, change the "Trial Status" to "Completed".
- (iii) Re-do step 6, i.e. click on the "Post-Trial" tab (see screenshot).



7. Under the "Post-Trial" tab, select the "Reports, Papers & Other Materials" subtab.
8. Under the "Reports & Other Materials" section, select "+ Upload a document" (see screenshot).
9. Upload your results report.
10. On the bottom of this page, click on "Register Changes" to save the edits.

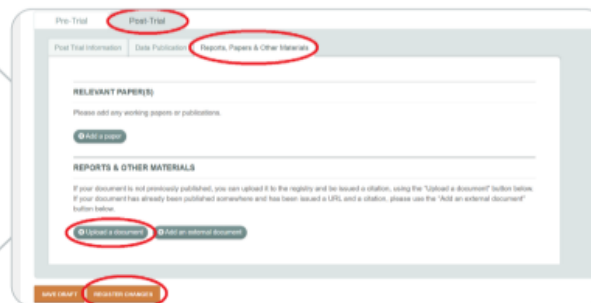


Figure A19: Complete Study Materials (10 pages)

Priors on Fraction Available And Fraction Null in RCTs in Economics

Start of Block: Block 1

[Intro] Our research team has read and encoded the registrations for a sample of studies on the American Economic Association Randomized Controlled Trial Registry (AEA Registry from now on) that were registered during 2015, 2016 and 2017. We extracted the main causal hypotheses for each study based on our reading of the primary outcomes and treatment arms as described in the registration.

End of Block: Block 1

Start of Block: Default Question Block



[num_h] What do you think is the median number of main hypotheses per study we extracted in the sample of registrations on the AEA Registry? (A note on our process: we generally did not encode heterogeneity or subgroup tests as main hypotheses unless they were explicitly emphasized as primary tests.)

Page Break

[any_res] For each encoded study, we investigated carefully whether the study had made publicly available any results in any of the following formats: a published article, working paper, book chapter, report, or dissertation. Approximately 8 to 9 years after registration on the AEA Registry, what percent of studies do you think have produced any publicly available results?

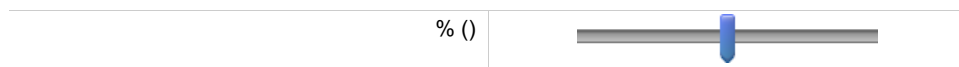
0 10 20 30 40 50 60 70 80 90 100



Page Break

[at_least_one] It is possible that even when results have been produced for a study, none of the available results correspond to the main hypotheses that were registered on the AEA Registry. (For example, the publicly available results may use a different primary outcome and not present any of the ones that were registered.) Among the studies that did produce any results (you answered $\frac{\$any_res}{TotalSum}$ %), what percent do you think **do not** report results for any of the registered main hypotheses 8-9 years after the study was registered?

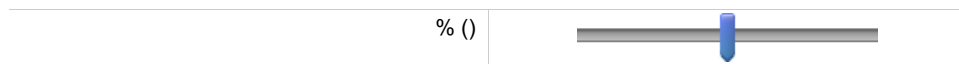
0 10 20 30 40 50 60 70 80 90 100



Page Break

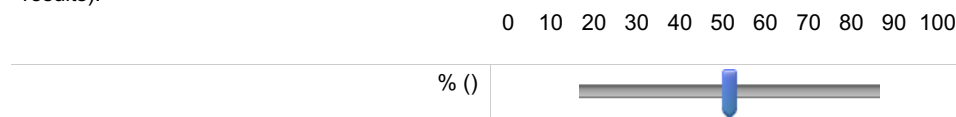
[Fraction available] Your responses imply that $\$e{\text{round}(\frac{\$any_res}{TotalSum} * (100 - \frac{\$at_least_one}{TotalSum}) / 100, 1)}\%$ of the registered studies have made publicly available results for **at least one** of the registered hypotheses ($\frac{\$any_res}{TotalSum}\% * (100\% - \frac{\$at_least_one}{TotalSum}\%) = \$e{\text{round}(\frac{\$any_res}{TotalSum} * (100 - \frac{\$at_least_one}{TotalSum}) / 100, 1)}\%$). Among the studies that do report results for at least one of the registered main hypotheses, what do you believe is the percent of registered hypotheses per study that is reported on average?

0 10 20 30 40 50 60 70 80 90 100



Page Break

[Fraction Null] Think about the same group of studies as the previous question (the $\text{round}(\frac{\text{any_res}}{\text{TotalSum}} * (100 - \frac{\text{at_least_one}}{\text{TotalSum}}) / 100, 1)\%$ that you estimated reported results for **at least one** hypothesis). Now focus on the registered hypotheses where a result was reported. What percent of these reported results per study do you think were null results (by which we mean a p-value ≥ 0.05) on average? For a made-up example, if a study registered 15 main hypotheses, reported results for 9 of them, and x of those were null results, then the percent of reported null results for this study would be $(x/9)*100$ (x of 9 reported results).



Page Break

[Summary] Here is a summary of your answers about what you think is publicly available from a study 8-9 years after registration: Percent of studies reporting any result: $\frac{\text{any_res}}{\text{TotalSum}}\%$ Percent of studies reporting results for at least one registered hypothesis: $\text{round}(\frac{\text{any_res}}{\text{TotalSum}} * (100 - \frac{\text{at_least_one}}{\text{TotalSum}}) / 100, 1)\%$ (= $\frac{\text{any_res}}{\text{TotalSum}}\% * (100\% - \frac{\text{at_least_one}}{\text{TotalSum}}\%)$) Average percent of hypotheses reported: $\text{round}(\frac{\text{Fraction available}}{\text{TotalSum}} * \frac{\text{any_res}}{\text{TotalSum}} / 100 * (100 - \frac{\text{at_least_one}}{\text{TotalSum}}) / 100, 1)\%$ (= $\frac{\text{Fraction available}}{\text{TotalSum}}\% * \text{round}(\frac{\text{any_res}}{\text{TotalSum}} / 100 * (100 - \frac{\text{at_least_one}}{\text{TotalSum}}) / 100, 1)\%$) Conditional on a result being reported: Average percent of hypotheses reported as null results: $\text{round}(\frac{\text{Fraction Null}}{\text{TotalSum}}, 1)\%$ of those studies that report at least one registered hypothesis (your $\text{round}(\frac{\text{any_res}}{\text{TotalSum}} * (100 - \frac{\text{at_least_one}}{\text{TotalSum}}) / 100, 1)\%$)

Please confirm or go back and revise your answers.

Page Break

[PAP] Comparing studies that have a Pre-Analysis Plan (PAP) in the AEA registry with studies that do not have one. Do you think that there will be systematic differences in: Average percent of hypotheses reported? Conditional on a result being reported: Average percent of hypotheses reported as null results?

	Percent with PAP is lower than percent without a PAP (1)	No systematic differences (2)	Percent with PAP is higher than percent without a PAP (3)
Reported hypotheses (1)	☐	☐	☐
Hypotheses reporting null results (2)	☐	☐	☐

[LMIC] Comparing studies from development economics, i.e., that focus on a study population in low and middle income countries (using the World Bank definition), with studies focusing on populations in high income countries. Do you think that there will be systematic differences in:

Average percent of hypotheses reported? Conditional on a result being reported: Average percent of hypotheses reported as null results?

	Percent for studies in LMIC is lower than percent not in LMIC (1)	No systematic differences (2)	Percent for studies in LMIC is higher than percent not in LMIC (3)
Reported hypotheses (1)	☐	☐	☐
Hypotheses reporting null results (2)	☐	☐	☐

[Published] Comparing studies that have been published in a peer-reviewed journal with studies that have not been published (yet) but have a manuscript available (i.e., a working paper, dissertation or report). Do you think that there will be systematic differences in: Average percent of hypotheses reported? Conditional on a result being reported: Average percent of hypotheses reported as null results?

	Percent in studies with publication lower than percent in studies with unpublished manuscripts (1)	No systematic differences (2)	Percent in studies with publication higher than percent in studies with unpublished manuscripts (3)
Reported hypotheses (1)	☐	☐	☐
Hypotheses reporting null results (2)	☐	☐	☐

[Top 5] Comparing studies that have been published in a top-5 economics journal with studies that have been published in other peer-reviewed journals. Do you think that there will be systematic differences in: Average percent of hypotheses reported? Conditional on a result being reported: Average percent of hypotheses reported as null results?

	Percent in studies with top-5 publication lower than percent in studies with other non-top-5 publication (1)	No systematic differences (2)	Percent in studies with top-5 publication higher than percent in studies with other non-top-5 publication (3)
Reported hypotheses (1)	☐	☐	☐
Hypotheses reporting null results (2)	☐	☐	☐

Page Break

End of Block: Default Question Block

Start of Block: Block 3

Q35 Have you ever posted a study in the AEA Registry?

- ☐ No (1)
- ☐ Yes (2)

Display this question:

If Have you ever posted a study in the AEA Registry? = Yes

Q36 If yes, what was the first year in which you posted on the Registry?

Q37 Have you seen a presentation by the research team on this study? The team is: Fernando Hoces de la Guardia, Edward Miguel, Viviane Silva da Rocha, Gufran Pathan, Erik O. Sørensen, Bertil Tungodden.

- ☐ No (1)
- ☐ Yes (2)

End of Block: Block 3

Start of Block: Block 2

Q39 This final section contains questions about your background, which are optional.

Q22 What describes you best?

- ☐ Graduate student (Master level or PhD) (1)
- ☐ Faculty (2)
- ☐ Post-doc (3)
- ☐ Non-academic researcher (4)
- ☐ Other (5)

Display this question:

If What describes you best? = Other

Q36 If other, please describe:

Display this question:

If What describes you best? = Graduate student (Master level or PhD)

Or What describes you best? = Faculty

Or What describes you best? = Post-doc

Q24 Discipline:

- ☐ Economics (1)
 - ☐ Political Science (2)
 - ☐ Psychology (3)
 - ☐ Sociology (4)
 - ☐ Other (5)
-

Display this question:

If Discipline: = Other

Q37 If other, please describe:

Display this question:

If What describes you best? = Graduate student (Master level or PhD)

Or What describes you best? = Faculty

Or What describes you best? = Post-doc

Q25 Title:

- ☐ Professor (1)
- ☐ Assistant Professor (2)
- ☐ Post-doc (3)
- ☐ Researcher (4)
- ☐ PhD Student (5)
- ☐ Masters Student (6)
- ☐ Associate Professor (7)
- ☐ Other (8)

Display this question:

If Title: = Other

Q38 If other, please describe:

Display this question:

If What describes you best? = Non-academic researcher

Or What describes you best? = Other

Q28 What type of organization do you currently work for?

- ☐ Academia/research institute (1)
- ☐ Government (2)
- ☐ International organization (3)
- ☐ NGO (4)
- ☐ Private sector (5)
- ☐ Other (6)

Display this question:

If What type of organization do you currently work for? = Other

Q29 If other, please describe:

Display this question:

If What describes you best? = Non-academic researcher

Or What describes you best? = Other

Q31 What is the highest level of education you have attained?

- ☐ Bachelor's degree (1)
- ☐ Masters degree (2)
- ☐ PhD (3)
- ☐ Other graduate/professional degree (4)

End of Block: Block 2

Table A1: Main Treatment Effects, With and Without the Baseline-*Available* Control

	Available (DD)		Available (CS)				Explained (CS)				Fraction Missing (CS)				Null (DD)		Null (CS)			
	main	pool	main		pool		main		pool		main		pool		main	pool	main		pool	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
Awareness	0.004 (0.003)	0.004 (0.003)	0.055 (0.066)	0.004 (0.007)	0.055 (0.066)	0.004 (0.007)	0.043 (0.021)	0.052 (0.023)	0.043 (0.021)	0.052 (0.023)	-0.099 (0.067)	-0.056 (0.025)	-0.099 (0.067)	-0.057 (0.026)	0.003 (0.002)	0.003 (0.002)	-0.012 (0.053)	-0.046 (0.034)	-0.012 (0.053)	-0.046 (0.033)
Engagement	0.069 (0.022)	0.069 (0.022)	0.095 (0.066)	0.066 (0.020)	0.095 (0.066)	0.066 (0.020)	0.119 (0.033)	0.123 (0.033)	0.119 (0.033)	0.123 (0.033)	-0.213 (0.067)	-0.190 (0.037)	-0.213 (0.067)	-0.190 (0.037)	0.054 (0.019)	0.054 (0.019)	0.084 (0.056)	0.066 (0.037)	0.084 (0.056)	0.066 (0.037)
Resources	0.055 (0.021)	0.055 (0.021)	0.136 (0.067)	0.056 (0.022)	0.136 (0.067)	0.056 (0.022)	0.069 (0.026)	0.083 (0.028)	0.069 (0.026)	0.083 (0.028)	-0.205 (0.067)	-0.139 (0.035)	-0.205 (0.067)	-0.139 (0.035)	0.029 (0.014)	0.029 (0.014)	0.093 (0.056)	0.041 (0.036)	0.093 (0.056)	0.041 (0.036)
Pooled (Eng. or Res.)	0.000 (0.015)	0.000 (0.015)	0.062 (0.015)	0.062 (0.015)	0.062 (0.015)	0.062 (0.015)	0.115 (0.059)	0.061 (0.015)	0.115 (0.059)	0.061 (0.015)	0.094 (0.022)	0.103 (0.024)	0.094 (0.022)	0.103 (0.024)	-0.209 (0.060)	-0.164 (0.028)	0.042 (0.012)	0.042 (0.012)	0.089 (0.050)	0.053 (0.031)
Post	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Available (baseline)			0.930 (0.022)	0.930 (0.022)	0.929 (0.021)	0.929 (0.021)	-0.156 (0.034)	-0.156 (0.034)	-0.158 (0.034)	-0.158 (0.034)	-0.774 (0.038)	-0.774 (0.038)	-0.771 (0.038)	-0.771 (0.038)			0.614 (0.039)	0.614 (0.039)	0.613 (0.039)	0.613 (0.039)
N	634	634	317	317	317	317	317	317	317	317	317	317	317	317	634	634	317	317	317	317
Study Fixed Effects	X	X													X	X				
Control			X	X	X	X	X	X	X	X	X	X	X	X			X	X	X	X
Baseline Avail control							X	X	X	X					X	X				
H1: Awareness	[0.073]		[0.200]	[0.268]			[0.019]	[0.012]			[0.069]	[0.013]			[0.091]		[0.591]	[0.915]		
H2: Eng. – Aware.	[0.001]		[0.253]	[0.002]			[0.024]	[0.025]			[0.027]	[0.000]			[0.004]		[0.022]	[0.000]		
H3: Res. – Eng.	[0.687]		[0.243]	[0.635]			[0.885]	[0.850]			[0.555]	[0.868]			[0.850]		[0.430]	[0.763]		

Note: Appendix companion to Table 4. Same outcomes and treatment arms; *Null* (Y_3 , fraction of pre-registered hypotheses with reported null results, $p > 0.05$) is treated like *Available* (both DD and CS, with and without baseline-*Available* control). Each cross-sectional specification is reported both *without* (left) and *with* (right) baseline *Available* (the pre-intervention value of Y_1) as a control. “Control” = pre-specified study-level controls: indicators for whether the study has a PAP, whether the study was undertaken in an LMIC, and indicators for each encoder on the team. Robust standard errors in parentheses; one-sided unadjusted p -values in square brackets. See Table 4 for the Romano-Wolf adjusted p -values from the pre-registered SUR family.

Table A2: Registration (G_0) Characteristics By Encoder

Dimension	All	E1	E2	E3
A: Registrations				
Number of Studies	317	136	139	31
Number of hypotheses	5,048	2,164	2,259	288
Registered with PAP	0.4	0.41	0.38	0.42
Has Published Paper	0.49	0.48	0.54	0.35
Has Published in Top 5 Journal	0.1	0.12	0.07	0.06
Studies populations in LMIC	0.65	0.65	0.65	0.61
Specifies heterogeneity in detail	0.12	0.12	0.15	0
B: Elements of the pre-registration				
Interventions (incl. control)	3.77	3.79	3.79	3.29
Arms	4.79	4.85	4.7	3.61
Main Outcomes	5.36	5.43	5.06	4.81
Hypotheses (Total) per Study	15.92	15.91	16.25	9.29
Main Hypotheses	11.77	11.82	11.54	9.29
Primary Heterogeneity Hypotheses	4.15	4.09	4.71	0
Has more than 10 Hypotheses	0.43	0.43	0.43	0.39
C: Fraction of hypotheses per registration that are				
Detailed (Overall)	0.31	0.27	0.34	0.35
Type of hypotheses: Simple Difference	0.92	0.93	0.9	1
Type of hypotheses: Double Difference	0.03	0.02	0.03	0
Type of hypotheses: 'Other'	0.05	0.05	0.07	0

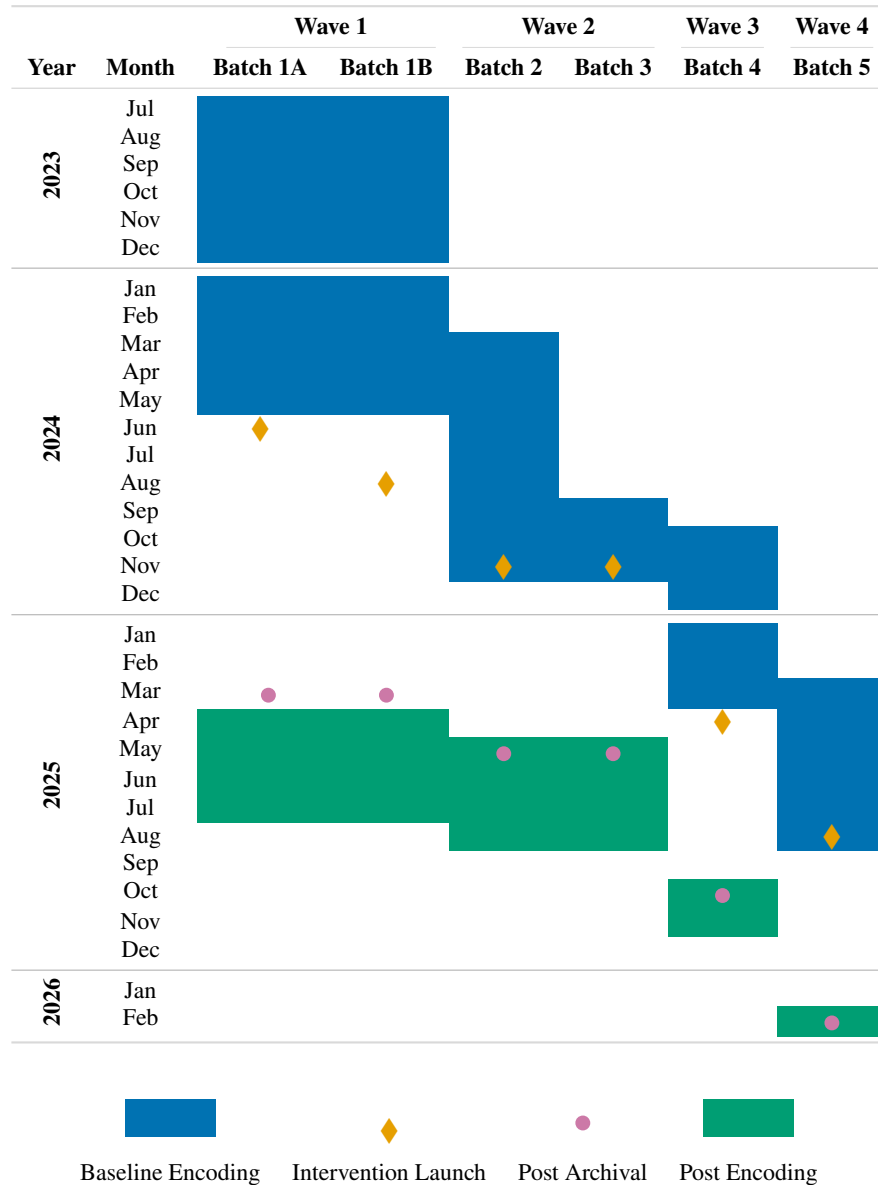
Note: Each column reports the same characteristics as Table 1 separately for studies encoded by each research assistant (E1, E2, E3) and for all studies combined. The table is intended to assess encoder balance: similar values across encoders suggest that the encoding assignment did not systematically differ across study characteristics. See Section 2.1 for details on the encoding process.

Table A3: Fraction Available and Fraction Null By Encoder

Categories	Available (%)				Null Available (%)				N			
	All	E1	E2	E3	All	E1	E2	E3	All	E1	E2	E3
A: Overall												
All Registrations	42.0	42.6	42.5	48.3	64.5	64.3	65.3	61.3	317	136	139	31
Registration with paper available	52.0	52.7	50.1	74.9	64.5	64.3	65.3	61.3	256	110	118	20
B: Type of publication												
Working paper	51.6	51.0	46.6	85.8	66.7	67.6	67.4	57.8	98	45	42	9
Published not top 5	55.1	57.1	53.0	66.0	65.5	64.1	66.4	68.1	125	49	65	9
Published top5	41.3	44.5	43.1	65.4	51.3	56.3	43.0	45.8	31	16	10	2
C: Stringency of availability												
In main body	34.8	35.9	33.6	44.2	61.1	62.7	59.5	59.9	317	136	139	31
No modifications	28.6	32.8	25.1	34.4	66.0	65.2	67.2	64.8	317	136	139	31
No mod. + main body	24.0	27.9	20.1	31.7	62.5	64.0	60.1	63.7	317	136	139	31
No mod + main body + not hard to find	19.7	18.1	20.1	31.6	64.3	68.5	60.1	66.8	317	136	139	31
Only main hypotheses	44.7	45.3	45.8	48.3	64.4	64.5	65.1	61.3	317	136	139	31
Study has heterogeneity:												
– All	31.4	42.1	24.0		66.0	68.3	62.7		39	16	21	0
– Main Only	53.8	64.6	45.5		65.4	69.5	60.4		39	16	21	0
– Heterogeneity Only	22.3	31.4	16.2		64.2	71.1	50.0		39	16	21	0
D: Other characteristics												
Studies have PAP	43.9	39.4	43.1	76.5	67.3	64.7	71.6	63.2	127	56	53	13
Studies in LMICs	37.8	40.0	37.7	40.0	64.3	62.5	65.3	71.0	206	88	91	19
Most hypotheses are clearly defined	43.0	42.0	43.2	61.9	67.4	60.8	73.7	61.9	100	36	47	11
More than 10 pre-registered hypotheses	31.8	35.4	31.6	26.0	68.8	69.4	71.1	53.1	137	58	60	12

Note: Each column reports the same categories as Table 2 separately for each encoder (E1, E2, E3) and for all studies combined. *Available* is the fraction of pre-registered hypotheses with a numerical result in any publicly accessible source; *Null | Available* is the fraction of available results with $p > 0.05$. The table documents that the two main encoders (E1, E2, accounting for 90.2% of studies) produce broadly similar results across most categories. E3 differences should be interpreted with caution given the small number of studies. See Section 2.1.

Figure A20: Timeline of RCT Study by Month (July 2023 – February 2026)



Note: This figure shows the timeline of baseline encoding, intervention rollout, and follow-up encoding by study batch. Intervention rollout dates by wave: Wave 1 on June 27, 2024 (Batch 1A, N = 12) and August 14, 2024 (Batch 1B, N = 128); Wave 2 on November 19, 2024 (Batches 2 & 3, N = 88); Wave 3 on April 2, 2025 (Batch 4, N = 77); Wave 4 on August 5, 2025 (Batch 5, N = 12). Wave sizes are analysis-sample studies per batch (sum = 317; the 35 archived control studies are excluded). When multiple events occur in the same month for a batch, the milestones (symbols) are overlaid on the encoding periods.

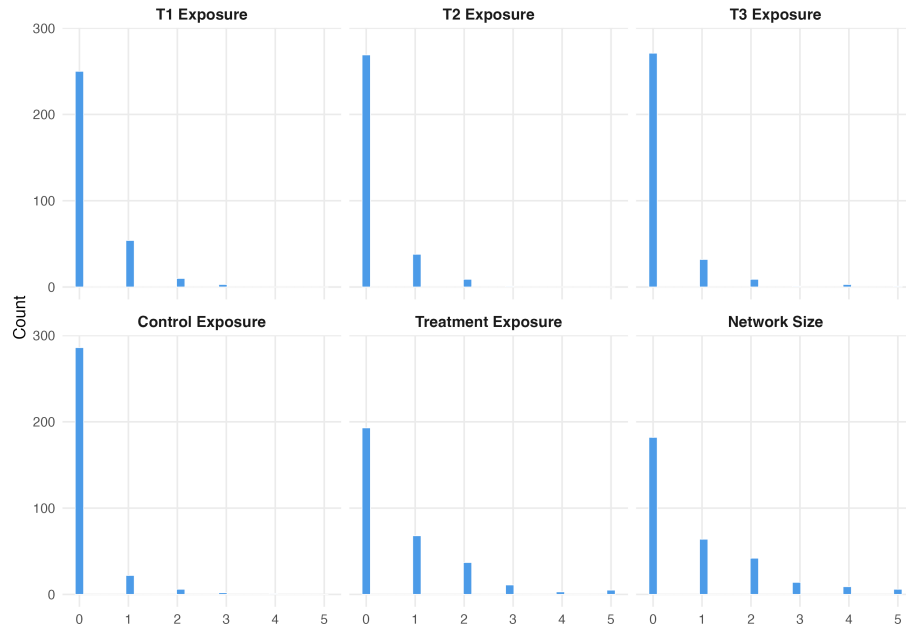


Figure A21: Distribution of Exposure Variables

Notes: Each panel shows the distribution of network-based exposure variables across studies, as defined in the pre-analysis plan. For study i , T_k -exposure ($k \in \{1, 2, 3\}$) denotes the number of co-authors on study i who are also co-authors on at least one other study assigned to arm T_k . Control exposure is defined analogously for the control arm. The spillover index $S_i = T_1\text{-exposure} + T_2\text{-exposure} + T_3\text{-exposure}$ captures a study's direct exposure to treated co-author links — the primary channel through which information or reporting norms could contaminate the control group or reinforce treatment effects. Network size $P_i = S_i + \text{Control-exposure}$ denotes the total number of co-author links to other studies in the sample, regardless of arm assignment.

Table A4: Spillover Analysis

	Available		Null	
	Add. Eq. 3	Add. Eq. 4	Add. Eq. 3	Add. Eq. 4
Post	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Awareness	0.001 (0.004)	0.001 (0.004)	0.002 (0.003)	0.002 (0.003)
Engagement	0.069** (0.029)	0.068** (0.029)	0.053** (0.026)	0.053** (0.026)
Resources	0.069** (0.028)	0.070** (0.029)	0.036** (0.018)	0.036* (0.019)
Observations	634	634	634	634
Study FE	X	X	X	X
Panel B: Spillover Parameters				
β_0 (spillover onto T_0)	0.000 (0.000)		0.000 (0.000)	
θ (complementarity)	0.009 (0.009)		0.008 (0.008)	
Panel C: Hypothesis Tests				
S-H2: $\beta_0 = 0$ (p-value)	1.000		1.000	
S-H3: $\beta_0 + \theta = 0$ (p-value)	0.330		0.296	
S-H4: $\theta > 0$ (one-sided p-value)	0.165		0.148	
S-H1: Any spillovers = 0 (Wald p, df)	0.703 [df=9]		0.781 [df=9]	

Standard errors clustered at the study level (in parentheses). Columns (1)-(2): Y1 = fraction of results available. Columns (3)-(4): Y3 = fraction of null results reported. Add. Eq. 3 (restricted) and Add. Eq. 4 (full arm-specific) are Addendum Equations 3 and 4 in the PAP addendum. S_i counts co-author exposures to any treatment arm; P_i controls for total network size. S-H2, S-H3 are two-sided; S-H4 is one-sided; S-H1 uses a chi-squared Wald test. * p<0.1, ** p<0.05, *** p<0.01.

Table A5: Spillover Analysis - Full Table

	Available		Null	
	(1) Add. Eq. 3	(2) Add. Eq. 4	(3) Add. Eq. 3	(4) Add. Eq. 4
<i>Panel A: Treatment effects</i>				
Awareness	0.001 (0.004)	0.001 (0.004)	0.002 (0.003)	0.002 (0.003)
Engagement	0.069** (0.029)	0.068** (0.029)	0.053** (0.026)	0.053** (0.026)
Resources	0.069** (0.028)	0.070** (0.029)	0.036** (0.018)	0.036* (0.019)
Post	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
<i>Panel B: Restricted spillover model (Add. Eq. 3)</i>				
Post $\times S_i$ (control)	0.000 (0.000)		0.000 (0.000)	
Post $\times S_i$ (treated)	0.009 (0.009)		0.008 (0.008)	
Post $\times P_i$	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Awareness $\times P_i$	-0.004 (0.006)	-0.001 (0.003)	-0.005 (0.006)	0.000 (0.003)
Engagement $\times P_i$	-0.007 (0.019)	-0.045** (0.020)	-0.006 (0.018)	-0.045*** (0.017)
Resources $\times P_i$	-0.026** (0.013)	-0.010 (0.014)	-0.015 (0.010)	-0.002 (0.011)
<i>Panel C: Arm-specific spillover model (Add. Eq. 4)</i>				
Post $\times S_{i,Awareness}$		0.000 (0.000)		0.000 (0.000)
Post $\times S_{i,Engagement}$		0.000 (0.000)		0.000 (0.000)
Post $\times S_{i,Resources}$		0.000 (0.000)		0.000 (0.000)
Awareness $\times S_{i,Awareness}$		0.005 (0.007)		0.003 (0.006)
Awareness $\times S_{i,Engagement}$		0.007		0.003

Continued on next page

Table A5 continued

	Available		Null	
	(1) Add. Eq. 3	(2) Add. Eq. 4	(3) Add. Eq. 3	(4) Add. Eq. 4
		(0.008)		(0.005)
Awareness $\times S_{i,Resources}$		0.003		0.001
		(0.004)		(0.003)
Engagement $\times S_{i,Awareness}$		0.100		0.110
		(0.070)		(0.068)
Engagement $\times S_{i,Engagement}$		0.052		0.030
		(0.042)		(0.027)
Engagement $\times S_{i,Resources}$		0.017		0.028
		(0.022)		(0.018)
Resources $\times S_{i,Awareness}$		-0.015		-0.006
		(0.010)		(0.008)
Resources $\times S_{i,Engagement}$		-0.012		-0.007
		(0.009)		(0.007)
Resources $\times S_{i,Resources}$		-0.004		-0.008
		(0.016)		(0.013)
Observations	634	634	634	634
Study FE	X	X	X	X

Standard errors clustered at the study level (in parentheses). Columns (1)–(2): Fraction of results available. Columns (3)–(4): Fraction of null results reported. Add. Eq. 3 (Addendum Equation 3) is the restricted spillover model; Add. Eq. 4 (Addendum Equation 4) is the full arm-specific model. S_{ik} denotes co-author exposure counts to arm k ; S_i aggregates across treated arms; P_i is total network size. Since there is no change in control group over time, the β_{jk} block in Add. Eq. 4 is not identified; the corresponding coefficients are zero. The covariance matrix of the 12 arm-specific spillover terms in Eq. 4 is therefore rank-deficient (rank 9); the S-H1 Wald test uses a Moore–Penrose pseudoinverse with $df = 9$. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A Details on Encoding Procedure

A.1 Encoding Protocol

This appendix describes the protocol used to encode pre-registration records (G0) and associated research outputs (G1) from the AEA RCT Registry. The protocol governs three stages: determining which studies enter the sample, extracting the relevant elements of each pre-registered design, and matching reported results back to pre-registered hypotheses. The rules were developed iteratively over the course of encoding several hundred studies and reflect recurring decisions made to handle ambiguity consistently and replicably.

Sample Inclusion

We restrict the sample to studies involving a genuine RCT in which treatment is assigned by the study team. Registrations are excluded when no treatment randomization occurs, when the entry is effectively empty, when results appear in the registry abstract (suggesting they post-date the analysis), or when randomization applies only to survey sampling. Studies confirmed registered only after data analysis are excluded. Studies described as pilots are retained when they implement a fully randomized design. A study registered after the intervention has concluded is retained only if the authors explicitly state that no outcome data had been analyzed prior to registration.

When a registration is modified after trial completion, the operative version is the last one predating the end of the intervention or the beginning of outcome data collection, unless the authors attest that later revisions were made without access to post-treatment data. When multiple PAPs exist, only the first one predating data collection for primary outcomes is used. If a referenced private PAP cannot be obtained despite outreach, encoding proceeds on public registry materials alone; if available only in a non-English language, machine translation is used. When a PAP and registry entry conflict, the PAP takes precedence if it predates data collection; if both predate the intervention, the registry entry is the public version of record.

G0 Encoding

We record populations, interventions, arms, outcomes, hypotheses, and pre-specified heterogeneity analyses. Multiple phases are treated as a single study, with later phases coded as additional populations. Missing or vague intervention descriptions are reconstructed from the abstract and design description. Any procedure that would not have occurred absent the study is treated as an intervention; non-randomized components are not.

Phase-in and stepped-wedge designs are treated as one treatment arm and one control arm. Arms are mutually exclusive. In factorial designs, all independently randomized combinations are distinct arms. In audit studies, we assume factorial unless clearly indicated otherwise; when ambiguous between factorial and parallel, we default to parallel. When interactions are clearly pre-specified, the interaction structure is encoded; otherwise, we encode interactions in 2×2 designs but restrict higher-order factorials to

parallel contrasts. In cascading designs, hypotheses isolate the marginal effect of each incremental component. When no explicit control group is defined, the most plausible reference group is inferred and documented; non-randomized comparison groups are not valid arms.

When a registration lists an impractically large number of primary outcomes, we prioritize those most central to the stated question. Outcome families are encoded as indices when a composite is explicitly defined; otherwise, components are recorded separately. Vague outcomes preserve the registry language with the coder’s interpretation documented. Mechanism and process outcomes are encoded only when central to the research question, causally identifiable, and explicitly pre-registered. When a registered outcome could correspond to multiple measurement forms, each distinct form is coded separately.

Hypotheses identify the estimand as precisely as possible. In parallel-arm studies, we encode only the minimal comparisons needed—usually each treatment arm relative to control—without adding contrasts or joint tests unless explicitly pre-specified. In difference-in-differences designs, the estimand is the treatment-by-post interaction. ITT is primary when both ITT and LATE are pre-specified. Quantile treatment effects are distinct from average effects. When multiple specifications are pre-specified, we encode the most central one. Contingent hypotheses are encoded once, with the contingency recorded and G1 results matched using the realized condition. Hypotheses are marked as *not detailed* when the registration does not identify a specific operationalization, contrast, or estimating equation. Non-experimental comparisons are not encoded. Spillover hypotheses are encoded only when explicitly pre-specified and central to the design. Heterogeneity analyses are included only when subgroup definitions are clear, central, and sufficiently pre-specified; they are not encoded in isolation from the main hypothesis. When the registration distinguishes “different effects” from “separate effects,” that distinction is preserved; otherwise, “different effects” is the default.

G1 Encoding

G1 encoding matches reported results to corresponding G0 hypotheses, recording every deviation. Reported outcomes are classified as *modified consistent* when the alternative operationalization could not plausibly have been chosen ex post, and *modified inconsistent* otherwise—applied to changes in transformations, functional form, and outcome construction. When several reported measures could correspond to the same underspecified registered outcome, we select the broadest or most aggregated measure; if equally plausible, one is selected at random and documented. When pre-registered outcomes are combined into a single reported index, the component most aligned with the index is recorded as modified consistent and remaining components are coded as *not found*. If both binary and continuous forms were explicitly pre-registered and the paper reports one, the hypothesis is treated as found and unmodified.

A hypothesis is *completely found* when the treatment effect can be recovered with no assumptions beyond what is explicitly reported—whether directly stated, recoverable by arithmetic, inferred from arm means, or read from figures. Standard errors are recovered from confidence intervals as width divided by 3.92; from arm-level statistics, as the square root of the sum of squared arm-level standard errors. Partially available

results are coded as *partially found*. We do not reconstruct results requiring structural assumptions or unreported econometric steps.

When multiple specifications are shown, we select the column most closely matching the G0 hypothesis; absent pre-specified covariates, we prefer the unadjusted specification. If one hypothesis maps to several variants, one is selected at random without regard to statistical significance. Covariate changes are consistent when involving pre-treatment variables and inconsistent when introducing post-treatment controls. Minor changes not substantively altering the estimand are not treated as modifications. When a registration implies an endline comparison and the paper estimates difference-in-differences, or when a binary outcome is reported with logit/probit instead of a linear probability model, the result is modified consistent. Explicitly acknowledged PAP deviations are recorded accordingly.

Judgment Calls

Coding decisions not fully resolved by the protocol are documented as judgment calls in the G0 or G1 records, describing the ambiguity, the decision taken, and the rule or precedent applied. The most common judgment calls concern registration timing, ambiguous arm structure, underspecified outcomes, specification choice, and matching vague pre-registered outcomes to reported measures.

B The Combinatorial Structure of Pre-Registered Hypotheses

A recurring theme in the descriptive results (Section 2.1) is the large number of hypotheses that a single pre-registration can contain. This appendix makes the underlying arithmetic explicit and contrasts it with researchers' expectations.

The number of distinct hypotheses (H) implied by a study design is, to a first approximation, a product of its constituent dimensions. Each study targets one or more populations or subgroups of interest (P); it administers a set of interventions (I), which are combined into experimental and control arms (A); it records several outcomes (O); it compares those arms through one or more contrasts (C); and it may pre-specify one or more dimensions of treatment-effect heterogeneity (D). Taken together, these dimensions imply a hypothesis count of roughly

$$H \approx P I A O C D.$$

Because these dimensions enter multiplicatively rather than additively, the hypothesis count grows quickly even when a study is modest on each individual dimension. Table 1 (Panel B) reports the relevant distributions: the median study pre-registers 9 hypotheses in total, but the distribution has a long right tail, with a mean of nearly 16.

This simple but important decomposition seems to be not well understood by economists. Figure A22 compares the distribution of expert forecasts of the number of hypotheses in a typical study (collected through the prediction survey described in Section 2.3) with the median observed in the registry data. The median forecast was only 4 hypotheses, less than half of the 9 found in the data, and 85% of all forecasts fell below the observed median. Researchers thus systematically expect far fewer pre-registered hypotheses than studies actually contain.

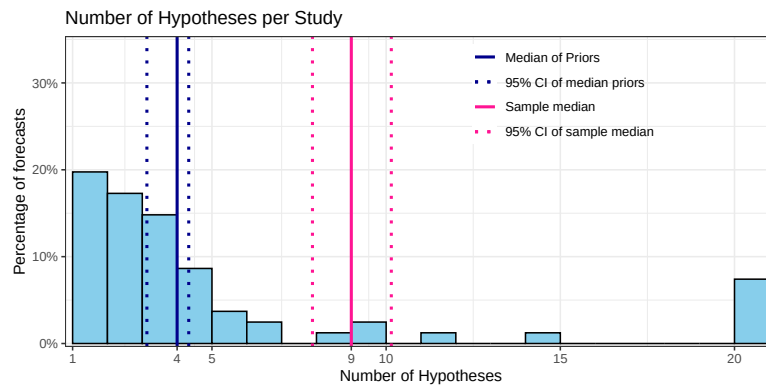


Figure A22: Distribution of expert forecasts of the number of pre-registered hypotheses per study (bars), with the median of the forecasts and the median observed in the AEA Registry data shown as vertical lines. 85% of forecasts fall below the observed median.

The same structure also points toward a remedy. Despite the combinatorial prolifer-

ation of tests, the overwhelming majority of pre-registered hypotheses can be expressed as a single comparison between two arms (Table A6): 89.1% are a simple difference, 4.5% a double difference, and only 6.4% fall into more complex categories. This suggests a practical principle for writing registrations: reduce quantity and increase clarity. Authors who pre-specify fewer, clearly prioritized contrasts, and who write each as an explicit two-arm difference, produce registrations that are easier to evaluate when registered and easier to follow through on later, consistent with the lower availability documented for studies with many (more than ten) pre-registered hypotheses in Section 4.1.

Table A6: Form of pre-registered hypotheses

Hypothesis form	Share of hypotheses
Single difference ($E[Y T_1] - E[Y T_0]$)	89.1%
Double difference	4.5%
More complex (e.g. joint test, structural estimate)	6.4%
Total	100%

Note: Shares are computed over all encoded hypotheses (main and heterogeneity); the three categories partition the set and sum to 100%.

C ML Prediction of Hypotheses and Paper (G_1) Availability

The descriptive results in Section 4.1 document that a substantial share of pre-registered hypotheses are never reported, but they are silent on whether this missingness is concentrated in particular kinds of hypotheses or studies. If observable features of the pre-registration alone can predict which results will go unreported, that would suggest the file drawer is at least partly systematic, and would identify *ex ante* the hypotheses most at risk of being lost. If, on the other hand, pre-registration features have little predictive content, the missingness pattern is closer to idiosyncratic, and broad interventions (such as those tested in the RCT) are more appropriate than targeted ones. This appendix reports an exploratory machine-learning exercise designed to distinguish between these cases.²³

We fit classifiers for two binary outcomes. The first is at the *hypothesis* level: whether a pre-registered hypothesis is *completely available* (Section 3). The second is at the *study* level: whether the study has *any* publicly available output containing results; a published article, working paper, report, presentation, blog or any other publicly available sources – which we denote as G_1 *available*. Predictors are constructed entirely from the pre-registration (G_0) data and from encoder metadata, with no information from the post-period. They include the number and length of pre-registered hypotheses, outcomes, interventions, and arms; characteristics of each hypothesis (test type, whether it is a main hypothesis, whether heterogeneity subgroups are pre-specified, whether control variables are listed); description-text lengths for hypotheses, outcomes, interventions, arms, and study populations; and the count and length of encoder *judgment calls* (free-text notes recorded when an encoding decision required interpretation), which we treat as a proxy for pre-registration ambiguity. Study-level features additionally include the existence of a PAP and an LMIC indicator.

We compare four classifiers spanning a range of flexibility: logistic regression (baseline), ℓ_1 -penalized logistic regression (LASSO), random forests, and gradient-boosted trees (XGBoost). All models are evaluated by 10-fold cross-validation, with hyperparameters tuned over a grid within each fold and selected by AUC. To prevent leakage across hypotheses from the same study, hypothesis-level folds are *grouped* by study, so that all hypotheses from a given study are held out together; study-level folds are stratified on the outcome. We report AUC, accuracy at the default 0.5 threshold, and accuracy/TPR/FPR/FNR at a fold-specific threshold that maximizes Youden’s $J = \text{TPR} - \text{FPR}$. Variable importance is extracted from the random forest (impurity), the gradient-boosted model (gain), and the LASSO (absolute coefficients), and is normalized to the maximum within each outcome–model combination.

²³This analysis is exploratory and was not pre-specified in the PAP. It is included as descriptive evidence on the structure of missingness, not as a hypothesis test.

Table A7: ML Feature Definitions

Construct	Hypothesis level	Study level
Hypothesis detail	Whether encoder judged description as detailed enough	Fraction of hypotheses judged detailed
Hypothesis type	Statistical test type (simple diff, DD, F-test, estimate)	Fraction that are simple difference-of-means
Hypothesis scope	Whether main hypothesis; whether it has heterogeneity subgroups	
Control variables	Whether control variables are specified	Fraction of hypotheses specifying controls
Arms	Number of treatment arms referenced	Total arms in the study
Interventions	Number of distinct interventions referenced	Total interventions in the study
Outcomes		Total outcomes encoded
Study size	Total hypotheses in the study	Total hypotheses; flag for > 10 hypotheses
Description lengths	Char length of outcome, hypothesis, detailed explanation, intervention, and arm descriptions	Study means of outcome, hypothesis, detailed explanation, intervention, arm, and population description lengths
Outcome/arm sharing	How many other hypotheses in the study test the same outcome or compare the same arms	
Complexity ratios		Outcomes per hypothesis; interventions per hypothesis
Study characteristics		Has a pre-analysis plan; LMIC study population
Encoder judgement calls		Count and total char length of encoder judgement calls, overall and by category (hypotheses, outcomes, arms, population)

All features derived from pre-registration (G0) data only. Chars = character length of description text.

Table A8: ML Model Performance Comparison

		At 0.5	At tuned threshold				
Model	AUC	Acc.	Thr.	Acc.	TPR	FPR	FNR
G1 Available							
GBM	0.57	0.81	0.85	0.56	0.50	0.18	0.50
	(0.07)	(0.00)	(0.01)	(0.06)	(0.08)	(0.06)	(0.08)
LASSO	0.61	0.81	0.81	0.60	0.55	0.20	0.45
	(0.02)	(0.00)	(0.01)	(0.05)	(0.07)	(0.05)	(0.07)
Logistic	0.57	0.79	0.83	0.57	0.52	0.23	0.48
	(0.03)	(0.01)	(0.03)	(0.05)	(0.07)	(0.09)	(0.07)
Random Forest	0.60	0.80	0.80	0.65	0.63	0.24	0.37
	(0.04)	(0.01)	(0.02)	(0.05)	(0.07)	(0.07)	(0.07)
Hypothesis Available							
GBM	0.62	0.67	0.31	0.63	0.70	0.43	0.30
	(0.02)	(0.02)	(0.03)	(0.03)	(0.08)	(0.08)	(0.08)
LASSO	0.63	0.67	0.34	0.68	0.59	0.27	0.41
	(0.02)	(0.02)	(0.01)	(0.02)	(0.06)	(0.05)	(0.06)
Logistic	0.61	0.66	0.32	0.61	0.74	0.46	0.26
	(0.03)	(0.02)	(0.02)	(0.03)	(0.05)	(0.05)	(0.05)
Random Forest	0.65	0.69	0.33	0.64	0.70	0.39	0.30
	(0.02)	(0.02)	(0.03)	(0.02)	(0.07)	(0.07)	(0.07)

Mean (SE) across cross-validation folds. AUC = Area Under the ROC Curve. Acc. = Accuracy. Thr. = classification threshold. TPR = True Positive Rate (sensitivity). FPR = False Positive Rate (1 - specificity). FNR = False Negative Rate (1 - sensitivity). Tuned threshold maximizes Youden's J (= TPR - FPR) per fold. CV = Cross-Validation. Feature definitions are given in the ML Feature Definitions table.

Variable Importance: Hypothesis Available

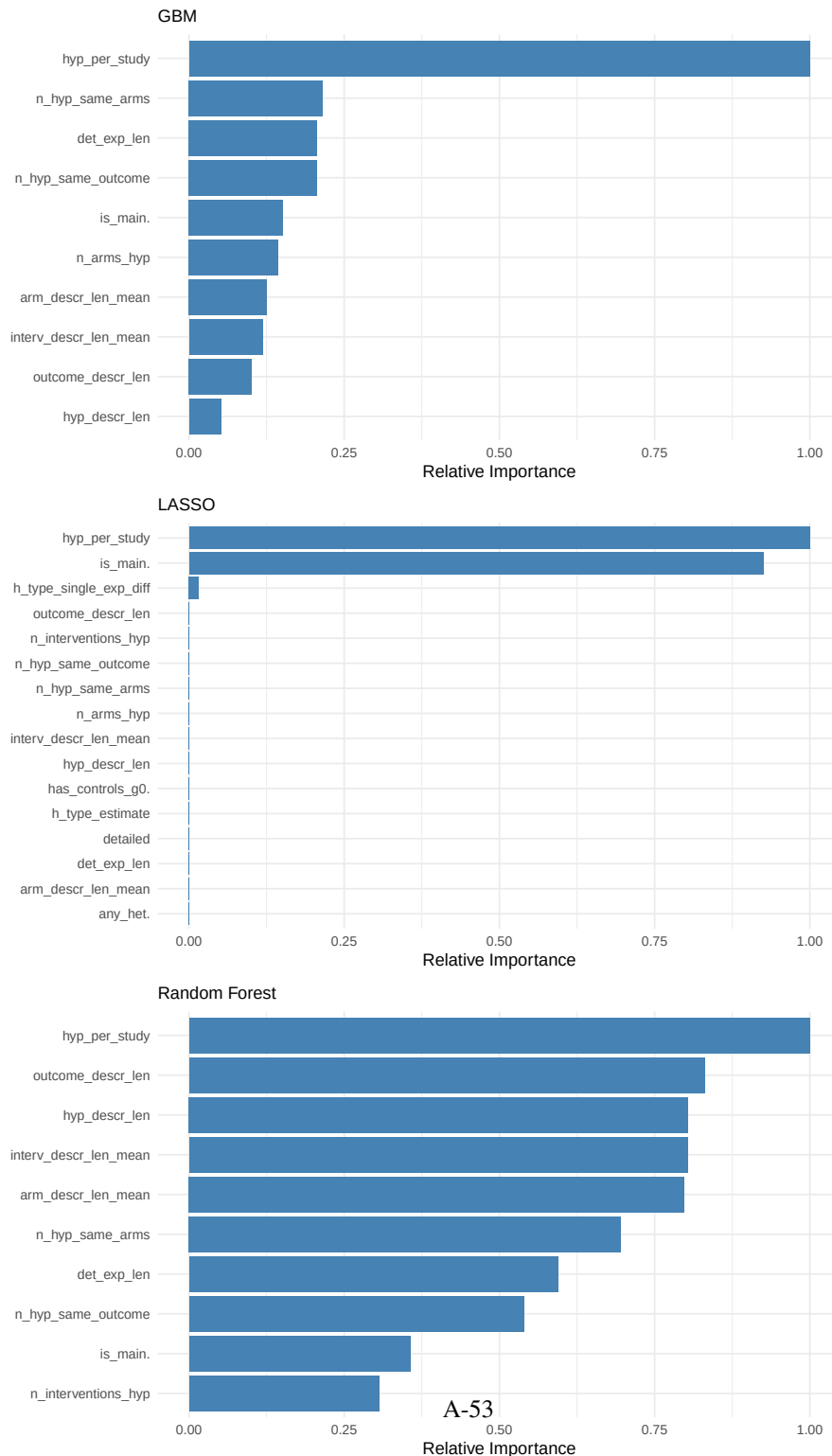


Figure A23: Variable Importance for Predicting Hypothesis-Level Availability

Note: Variable importance for predicting whether a pre-registered hypothesis is *completely available* (Section 3). Importance is extracted from the gradient-boosted model (center panel), the LASSO (absolute coefficient, middle panel), and the random

Variable Importance: G1 Available

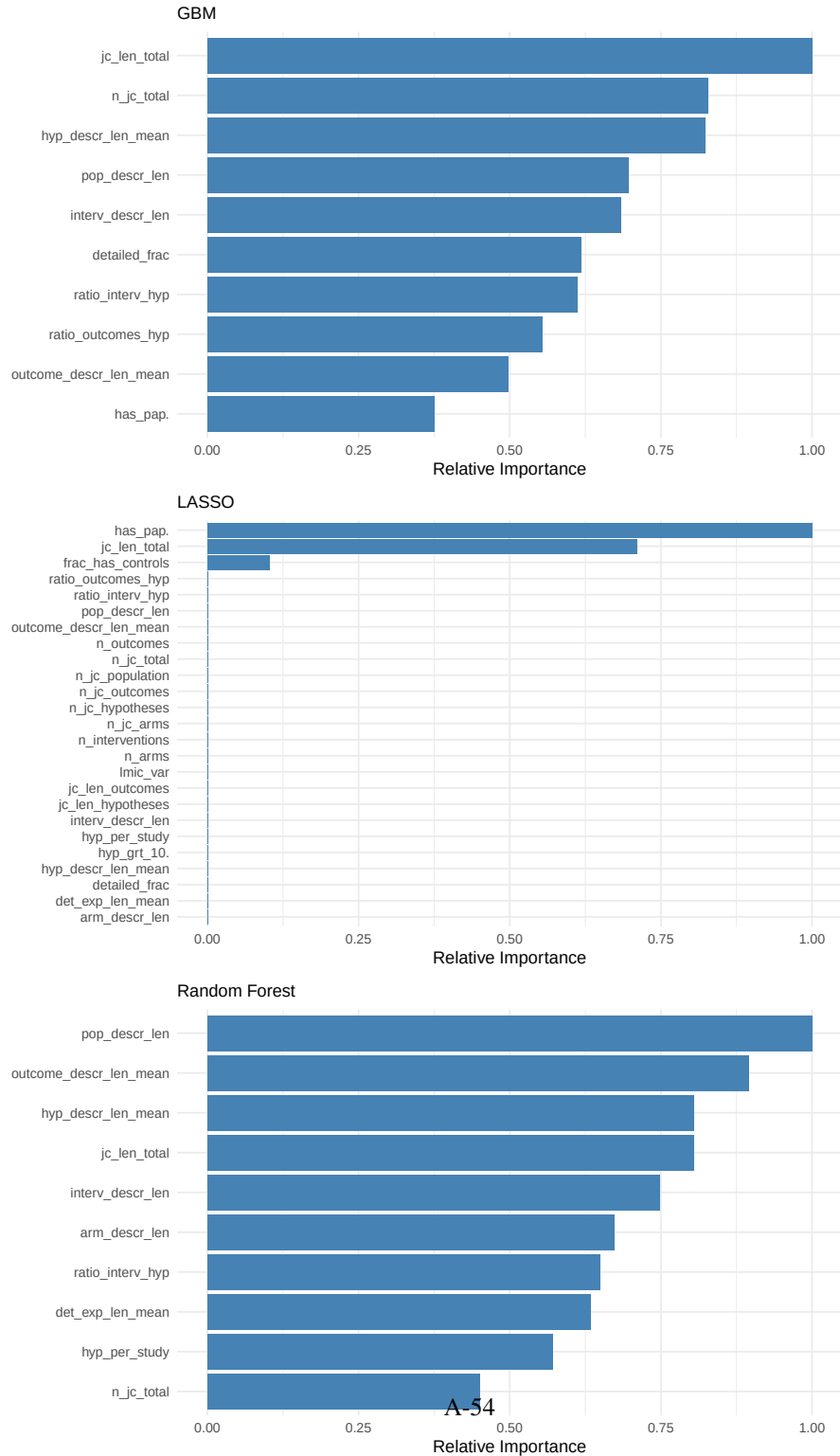


Figure A24: Variable Importance for Predicting Study-Level G_1 Availability

Note: Variable importance for predicting whether a study has any publicly available output containing results (G_1 available). Importance is extracted from the gradient-boosted model (gradient boost), the LASSO (least absolute shrinkage and selection), and